# Rethinking GPR: the Sources of Geopolitical Risk*

Irma Alonso-Álvarez
Banco de España

Marina Diakonova
Banco de España

Javier J. Pérez
Banco de España

José Nicolás Rosas
Banco de España

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**Abstract**

We develop a new measure of geopolitical risk (GPR) that decomposes aggregate risk into its geographical sources. Using news-based text analysis, we construct directed GPR indices that trace tensions back to their origin countries or regions, and we address the overlap across narratives by distinguishing between exclusive and non-exclusive components of risk. This outward-looking classification delivers a more detailed view of geopolitical tensions than existing aggregate indices and standard country-specific measures. Using structural VARs, we show that the macroeconomic effects of GPR depend on its geographical origin, with substantial differences in the magnitude, timing, persistence, and, in some cases, sign of the responses across shocks. A "No-War" counterfactual for the 2026 Iran conflict further illustrates that accounting for the geographical source of risk can materially sharpen the estimated macroeconomic effects of major geopolitical events.

**Keywords:** geopolitical risk; geopolitical tensions; textual analysis.
**JEL codes:** C43; E32; F51; F52; H56.

*Corresponding author: José Nicolás Rosas (email: josenicolas.rosas@bde.es). The authors would like to express their gratitude to Corinna Ghirelli, Alessandro Franconi, Matteo Iacoviello, Matthias Rottner, Dario Caldara, and participants at Banco de España's internal seminar series, PSE Macro Days 2024, and the European Commission's Annual Research Conference 2024, for their insightful comments and suggestions. The views expressed in this paper are those of the authors and do not necessarily reflect those of the Banco de España (BdE), the European Central Bank or the ESCB.

# 1 Introduction

Recent global events, including Russia's invasion of Ukraine in February 2022 and the outbreak of war in Iran in February 2026, have renewed interest in the role of geopolitical developments in macroeconomic fluctuations. Policymakers and analysts increasingly emphasize geopolitical risk (GPR) as a key determinant of economic activity over the short and medium term (e.g., Lagarde, 2024; Powell, 2023; Georgieva, 2024). Measuring this risk is therefore essential to understanding its economic consequences.

Caldara and Iacoviello (2022, hereafter C&I) provide a benchmark framework for measuring GPR, which they define as the *"threat, realization, and escalation of adverse events associated with wars, terrorism, and any tensions among states and political actors that affect the peaceful course of international relations"*. They operationalize this definition through a dictionary-based index measuring the share of articles covering adverse geopolitical events in leading English-language newspapers. The resulting GPR index has been widely used to study the macroeconomic and financial effects of geopolitical risk shocks (e.g., Brignone et al., 2026; Caldara et al., 2026; Georgiadis et al., 2024; Pinchetti, 2025, among others).

C&I also construct country-specific indices that capture geopolitical risks involving individual countries. Like the aggregate index, however, these measures reflect the perspective of the English-language newspapers on which they are based. At the same time, a growing literature has highlighted the importance of local-language sources for capturing domestic perceptions of geopolitical risk (Bondarenko et al., 2024; Ambrocio et al., 2025; Bondarenko et al., 2025).

We build on this literature and propose a decomposition of GPR into its geographical sources. We introduce the concept of *directed GPR*, which captures the risks a country perceives as originating from specific foreign countries or regions. For example, geopolitical risks perceived in the United States may originate in the Middle East, Russia, or China, with potentially distinct economic implications. Using local-language news sources, we construct directed GPR indices for four major economic regions: the United States, the EU (we consider the *big four*: Germany, France, Italy, and Spain), China, and Russia. We make our updated monthly indices available for download, as well as a more broad database of 34 such local GPRs[1]. These indices allow us to trace the perceived origins of geopolitical risk and distinguish between shared and country-specific components of outward-looking risk.

[1]Available for download at: `General and Bilateral Geopolitical Risk (GPR) Indices for 34 Economies`

Our decomposition first reveals that a substantial share of GPR coverage cannot be attributed to a specific external region. This *internal* component accounts for, on average, 46% of GPR articles in Russia and 37% in the United States[2]. These articles often discuss the domestic repercussions of geopolitical events or refer to foreign regions without sufficient focus to qualify for geographical attribution. An aggregate GPR index thus combines outward-looking concerns with domestic narratives surrounding geopolitical developments. Given our focus on the directed dimension of risk, we decompose the remaining *external* component into mutually exclusive categories associated with individual regions or combinations of regions.

The resulting directed indices reveal distinct patterns of geopolitical concern that are obscured by country-level aggregates. Both the relative importance of external regions and its evolution over time differ markedly across countries. In the United States, the share of external GPR focused exclusively on the Middle East falls from about 49% in the 2000s to about 32% from 2020 onward, while the corresponding focus on Russia and China rise from about 3% to 14% and from about 3% to 6%, respectively. China's external risk narrative, by contrast, remains centered on the Western Bloc, which alone accounts for about 24% of external coverage in 2020–2026. In Russia, the share devoted exclusively to neighboring countries rises from about 11% in the 2000s to about 32% in 2020–2026, overtaking the Western Bloc. These shifts show that changes in aggregate GPR can conceal substantial changes in its geographical composition, motivating a distinction between risks associated with different external sources.

Coverage involving multiple regions is also quantitatively important. In each of the four economic regions, a combination of focal regions ranks among the six largest components of external GPR. For Europe, for instance, China accounts for only about 2% of external coverage from 2020 and onwards when considered exclusively, but a further 7% when it appears jointly with other regions; the Western Bloc accounts for 21% alone and 48% once joint coverage is included. Simply summing directed indices would therefore count a substantial volume of coverage more than once. Our decomposition avoids this duplication while preserving narratives involving several regions, allowing us to distinguish changes in attention to a particular region from changes in coverage linking it to other geopolitical actors.

Finally, we demonstrate the value of our directed GPR indices through a number of empirical applications. First, we revisit the evidence on the macroeconomic effects of geopolitical risk using structural Bayesian vector autoregressions for the United States, China and Russia. We adjust the models for the COVID-19 period, applying the COVID

[2]For common samples, starting in January 2003.

volatility correction in Lenza and Primiceri (2022), allowing us to combine a long pre-pandemic sample with recent geopolitical events within a consistent framework. We find substantial heterogeneity in the macroeconomic effects of bilateral GPR shocks across geographical sources. While most shocks are followed by lower economic activity, the magnitude, timing, persistence, and, in some cases, sign of the responses vary depending on where geopolitical risk originates. The effects on prices and interest rates likewise show a considerable degree of heterogeneity. These results complement the existing evidence on the macroeconomic consequences of geopolitical risk (Bondarenko et al., 2024, 2025; Caldara and Iacoviello, 2022; Caldara et al., 2026; Pinchetti, 2025; Reynolds and Lloyd, 2026) and show that aggregate GPR measures can mask economically relevant differences in the transmission of risks originating in different regions.

Next, we quantify the macroeconomic cost for the U.S. of the recent escalation of the conflict in Iran through a counterfactual exercise. This exercise provides an estimate of the dynamics that would have prevailed absent the increase in geopolitical risk associated with the onset of the hostilities. Holding the aggregate U.S. GPR index at its February 2026 level, we estimate that by mid-2026 real GDP is approximately 0.3% lower and consumer prices 0.1% higher than under the counterfactual path. Since the geographical source of the tensions is known in this episode, we repeat the exercise using the directed U.S.–Middle East GPR index. The directed measure records a substantially larger increase in geopolitical risk and implies a larger decline in real GDP, of approximately 0.4%. This application illustrates how directed GPR indices complement aggregate measures by isolating the geographical source of risk and refining the estimated macroeconomic effects associated with specific geopolitical episodes.

We make all measures presented in this paper available through a regularly updated companion webpage. Our directed indices complement existing measures in the literature by capturing geopolitical risk from local perspectives and distinguishing its geographical sources. These series provide researchers, policymakers, and market participants with additional tools to monitor evolving geopolitical concerns and assess their economic implications.

Our paper relates to the growing field of geoeconomics, which examines the interplay between geopolitical forces and economic outcomes (see Mohr and Trebesch, 2025 for a survey). More specifically, it contributes to research on the measurement and macroeconomic effects of geopolitical risk, pioneered by Caldara and Iacoviello (2022). Within this literature, Bondarenko et al. (2024) emphasize the importance of a local perspective for capturing a country's or region's underlying perceptions of geopolitical risk.[3] Related

[3]The importance of differences in risk perceptions is also emphasized in the theoretical geoeconomics

work documents how these perceptions differ across news sources. Ambrocio et al. (2025) find that a Finnish GPR index based on local news diverges sharply from C&I's country-specific index, while Bondarenko et al. (2025) construct a EU index that captures regional perceptions and yields substantially stronger estimated macroeconomic effects than the English-language benchmark. We extend this literature by broadening the country coverage of GPR indices based on local news sources,[4] separating these indices into internal and external components, and further decomposing the external component into directed measures associated with individual foreign regions or combinations of regions.

Second, our paper contributes to the measurement of the directed dimensions of geopolitical risk and geoeconomic pressure. Caldara and Iacoviello (2022) construct country-specific indices that capture risks posed by, or involving, individual countries from the perspective of U.S. newspapers. Iacoviello and Tong (2026) extend this approach by using large language models to identify actors and their roles in geopolitical events, constructing directed indices that distinguish between the initiators and respondents of tensions. Closely related, Clayton et al. (2025) use large language models to extract measures of geoeconomic pressure from corporate earnings calls and analyst reports, identifying which governments exert pressure on which foreign targets and through which policy instruments. Other recent contributions use large language models to refine the measurement of geopolitical risk and uncertainty. Audrino et al. (2024) employ large language models to construct news-based GPR indices, while Ghomi and Hurtado (2026) combine semantic retrieval with LLM-based uncertainty scoring to measure geopolitical uncertainty. Ioannou et al. (2026) use multilingual European newspapers to distinguish geopolitical from geoeconomic tensions and identify the underlying sources of the latter, including trade, energy, finance, and technology. We complement these approaches by measuring the geographical sources of geopolitical risk from the perspective of each country's domestic news coverage. To the best of our knowledge, we construct the first set of directed GPR indices using each country's local newspapers rather than a common set of U.S. newspapers. Our contribution thus combines geographical disaggregation with variation in the national perspective from which risk is measured, capturing concerns associated with individual foreign regions as well as narratives involving multiple regions.

The rest of the paper is organized as follows. Section 2 describes the construction of our local-language GPR indices and compares them to various benchmarks, including

framework of Clayton et al. (2026), which distinguishes between hegemonic and non-hegemonic countries.

[4]To the best of our knowledge, we construct the first bilingual GPR index for China, drawing on both English- and Chinese-language news sources.

the indices by Caldara and Iacoviello (2022) and Bondarenko et al. (2024, 2025), focusing on the cases of the U.S., the EU, China, and Russia. Section 3 introduces the notion of *directed* GPR and showcases their usefulness in several ways. Section 4 presents a series of empirical applications: a decomposition of *directed* geopolitical risk to obtain national GPR profiles, an assessment of the macroeconomic effects of *directed* GPR shocks, and a "No-War" counterfactual exercise. Finally, Section 5 concludes.

# 2 Country-specific, local-language GPRs

## 2.1 Building country-specific local-language GPR indices

Geopolitical risk is inherently shaped by national context. While the original GPR index of Caldara and Iacoviello (2022) offers a valuable benchmark, it primarily reflects a U.S.-centric or "Anglo-Saxon" perspective, as most of its source newspapers are based in the United States. As emphasized by the authors themselves, this limits the index's ability to capture how geopolitical tensions are perceived in other parts of the world. In contrast, Bondarenko et al. (2024) show that using local-language sources can yield GPR indices that better reflect domestic narratives and have stronger explanatory power for local economic outcomes.

Motivated by this, we construct country-specific GPR indices using local-language newspapers for four major economic regions: the United States, the EU (which we obtain as a combination of the indices for the *big four*: Germany, France, Italy, and Spain), China, and Russia. Table 1 lists the sources used for each constituent country, as well as the starting date for which the corresponding index becomes available[5]. The search terms employed in each language are detailed in Appendix A.

Our methodology follows a standard dictionary-based approach. For each country, we count the number of articles from selected sources that contain at least one term from our geopolitical risk dictionary. This count is normalized by the total number of articles from the same sources in the same period, yielding a time-varying index of attention to geopolitical risk. To facilitate cross-country comparisons, we further normalize each index by its historical mean and scale it to 100.[6]

We also perform a narrative validation of our indices, to ensure that the peaks do not

[5]Which corresponds to at least two sources being present.

[6]When computing the index for the EU, we first ensure a common starting date from which all four countries become available (1997), normalize each individual index until 2019, and then create a sum weighted by the gross domestic product based on purchasing-power-parity (PPP) share of world total, 2023 figures (WEO dataset, as of April 2025).

Table 1: Newspaper sources and starting period

| |
|---|
| **C&I sources** |
| The Wall Street Journal, The Washington Post, The Los Angeles Times, The Chicago Tribune, The New York Times, USA Today, The Daily Telegraph, The Guardian, The Globe and Mail, The Financial Times |
| **This work´s sources (from Factiva)** |
| **United States (1985):** The Wall Street Journal, The Washington Post, The Los Angeles Times, The Chicago Tribune, The New York Times, The Boston Globe, USA Today |
| **China (1995):** *English*: South China Morning Post (HK), China Daily *Simplified Chinese*: Headline Daily, Oriental Daily News, Hong Kong Economic Journal, Sing Tao, Ming Pao, People's Daily, China Times |
| **Russia (1999):** Kommersant, Vedomosti, Novaya Gazeta, Argumentu y Faktu |
| **Germany (1997):** Süddeutsche Zeitung, Die Welt, Handelsblatt, Der Tagesspiegel, Die Tageszeitung, Börsen-Zeitung, Berliner Zeitung, Frankfurter AZ |
| **France (1995):** Le Figaro, Les Echos, Le Monde, Liberación, La Croix, L'Opinión, La Tribune, L'AGEFI Quotidien, Le Parisien |
| **Spain (1996):** La Vanguardia, El País, Cinco Días, El Mundo, 20 minutos, El Confidencial, ABC, El Periódico, El Economista, Expansión, El Español |
| **Italy (1997):** Corriere della Sera, La Stampa, Il Fatto Quotidiano, La Repubblica, Il Sole 24 Ore, Italia Oggi, Il Giornale |

correspond to false negative signal. In Appendix C this is shown for the U.S., China, and Russia. The validation of our European indices was done as part of work behind Alonso-Alvarez et al. (2026), and can be found therein[7].

## 2.2 Comparison with related indices

To firmly anchor our approach within the existing literature, we begin by comparing our local-language, Factiva-based GPR indices with the benchmark index of Caldara and Iacoviello (2022) (C&I), based on Anglo-Saxon sources. This comparison ensures that any differences we observe stem from deliberate changes in source language and coverage, rather than from methodological inconsistencies.

To this end, we first replicate the C&I index using Factiva. We apply the same dictionary-based methodology described in the previous section, using the English-language sources listed in Table 1. While perfect replication is not possible—due to differences in Fac-

[7]The EU indices are part of the `General database of local GPR indices`, released by the Bank of Spain that includes GPR indices for 34 countries, as well as four directed indices for each.

tiva's data coverage,[8] search engine behavior,[9] and technical constraints,[10]—our replication performs well across several dimensions.

First, the replicated index is highly correlated with the original (correlation = 0.82; see Figure B.1). Second, it captures nearly all major geopolitical events identified by C&I (Table B.1). Third, when embedded in the original VAR model of Caldara and Iacoviello (2022), our index produces nearly identical impulse response functions (Figure B.2). These results validate our Factiva-based replica as a reliable benchmark. Throughout the paper, references to the "C&I index" refer to this replication unless otherwise noted. See Appendix B for more details.

We now turn to the comparison between our local-language GPR indices and the replicated Anglo-Saxon benchmark. Figure 1 presents the results. While the indices share broadly similar dynamics, they also exhibit meaningful differences. Pair-wise correlation coefficients range from 0.99 for the U.S. and 0.92 for China to 0.84 for France and 0.67 for Russia with the Anglo-Saxon benchmark. Beyond correlations, the indices also differ in the timing and intensity of peaks: local indices are naturally more sensitive to region-specific developments. Appendix C presents additional details regarding the local narratives for each region.

In addition to the original C&I GPR index, several related measures are sometimes used as proxies for country-level geopolitical risk. The first are C&I's "country-specific" indices. Although the authors emphasize that these measures reflect an Anglosphere perspective on the geopolitical risks involving each country, they are nevertheless often used as proxies for that country's own perspective. These indices are constructed by adding terms associated with the focal country to the general GPR search query while retaining the same English-language news sources used for the original C&I index. We also compare our indices with the local-source measures of Bondarenko et al. (2024, 2025), which, like ours, are constructed using the Factiva database but differ in the specific news sources and translated search terms employed.

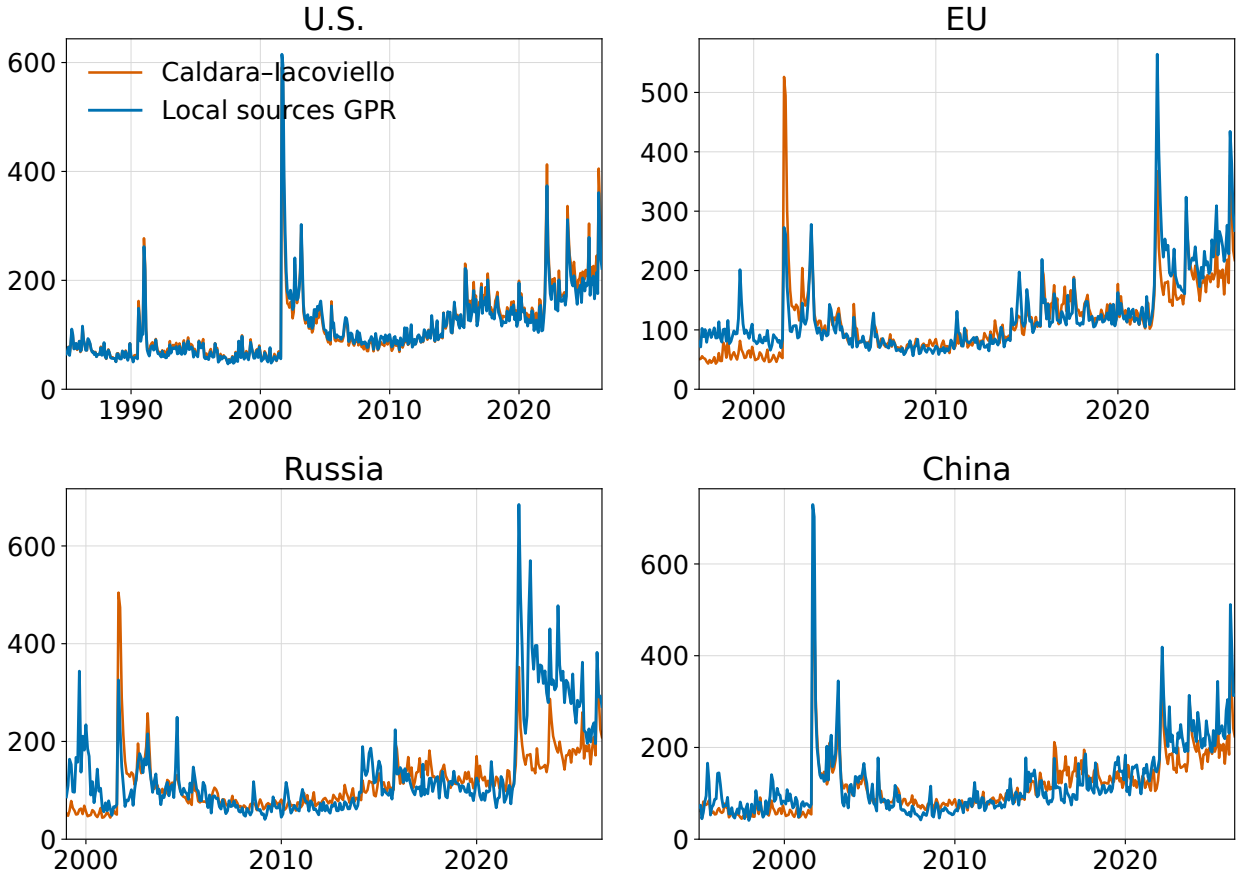
Figure 2 reports the pairwise correlations, with the final two rows showing the mean correlations across countries, including and excluding the EU aggregate. Our GPR indices are most strongly correlated with the local-source indices of Bondarenko et al. (2024) and Bondarenko et al. (2025). This is reassuring, given the similarities in the underlying data

[8] Two C&I sources are excluded: the *Financial Times* is not available in Factiva, and the *Daily Telegraph* contains excessive duplicates.

[9] Factiva does not disclose whether it uses lemmatization or stemming, which may affect article counts.

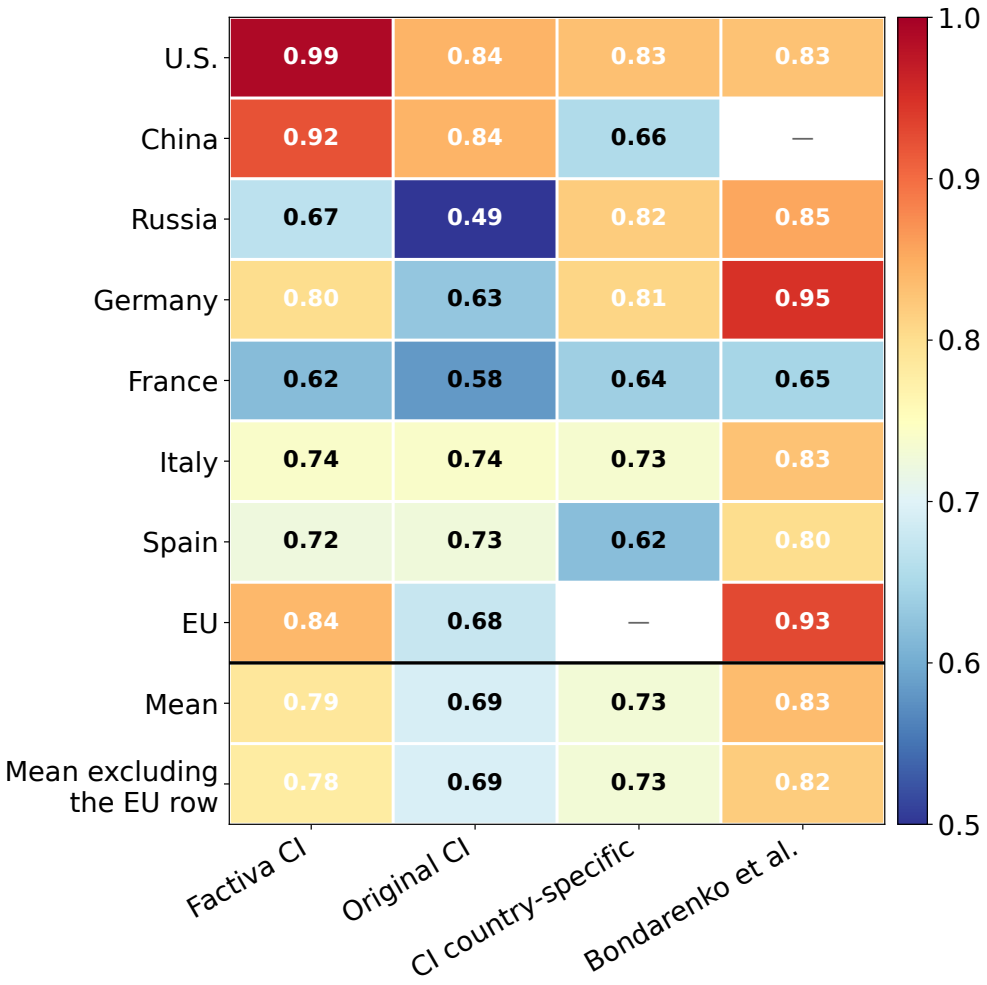
[10] Factiva's API limits query length, preventing us from excluding certain article categories (e.g., sports, entertainment). This issue is more pronounced in languages with longer word forms; see Section F.1 for details.

Figure 1: Different perspectives on GPR: local language newspapers versus Anglo-Saxon newspapers (C&I).




*Notes:* Each panel compares the corresponding local GPR index, constructed from the country's local news sources (blue), with our Factiva-based replication of the aggregate Caldara and Iacoviello (2022) GPR index using English-language newspapers (orange). The orange benchmark is identical across panels. Each index measures the share of news articles covering geopolitical risk and is normalized to a mean of 100 over the common observation period between the two indices. Differences between the series illustrate how the timing and intensity of geopolitical concerns vary with the perspective of the underlying news sources.

Figure 2: Correlations between GPR indices




*Notes:* Pairwise Pearson correlations between our local-source GPR indices and alternative GPR measures at monthly frequency. "Factiva C&I" denotes our reproduction of the aggregate Caldara and Iacoviello (2022) GPR index using Factiva and the same English-language news sources. "Original C&I" denotes the original aggregate GPR index of Caldara and Iacoviello (2022), as provided by the authors. "C&I country-specific" denotes the corresponding country-specific GPR index for each country, constructed by conditioning the C&I search on references to the focal country while retaining the original English-language news sources. "Bondarenko et al." denotes the local-source GPR indices constructed by Bondarenko et al. (2024) and Bondarenko et al. (2025), with the corresponding local measure used for each country or region. The final two rows report mean correlations across available comparisons, first including and then excluding the EU aggregate. Dashes indicate that the corresponding index is unavailable.

and construction methodology, and provides an external validation of our measures. Our indices are also more strongly correlated, on average, with our Factiva reproduction of the C&I index than with the original C&I series. This highlights the role played by both the underlying news database and the implementation of the search methodology.

Finally, although the C&I country-specific indices are, on average, more strongly cor-

related with our GPR measures than the original aggregate C&I index, important differences remain. This suggests that conditioning on a focal country captures some country-relevant variation, but does not reproduce the perspective embedded in local news coverage. The contrast is particularly pronounced for Russia and Europe, where relying on the C&I country-specific indices may provide a distorted picture of the geopolitical risks emphasized in domestic narratives. This further underscores the importance of local sources. In the next section, we combine this local perspective with a geographical decomposition of risk to measure how country $A$ perceives geopolitical risks originating from country or region $B$.

# 3 Directed GPRs

Having established local-language GPR indices that reflect country-specific narratives, we now turn to identifying the geographical origin of these risks. Specifically, we aim to quantify the risk perceived by a country as being associated to a particular external region. An important application of our methodology will then be to determine which external regions contribute most to the geopolitical risk experienced by each country.

First, we develop a method to associate local GPR indices with specific external regions[11]-—what we refer to as *directed GPRs*. These indices capture the part of a country's GPR that is attributable to tensions involving a particular region (usually foreign). Comparing a country's directed GPRs across such regions allows us to rank their relative importance in shaping a country's geopolitical risk profile.

Our approach builds on the dictionary-based method used to construct the local GPR indices. To focus on the contribution of a specific region $j$ to the GPR of country $i$, we augment the original query with terms or filters that identify articles focused on region $j$. The resulting directed GPR index reflects the frequency of such articles, normalized by the total number of articles from country $i$'s sources. In Caldara and Iacoviello (2022), similar indices are referred to as "country-specific" GPRs (as mentioned above). However, since all their indices are derived from Anglo-Saxon sources, they reflect a single narrative perspective. In contrast, our directed indices are constructed from the local media of each country, allowing us to distinguish both the origin ($i$) and the focal point ($j$) of the perceived risk.

Rather than relying solely on keyword mentions, we use Factiva's proprietary regional classification system (the "*re*" filter) to identify the geographical focus of each article. Al-

[11]We use the term "region" rather than "country" to allow for broader or more flexible geographical groupings, including areas defined by geopolitical or cultural boundaries.

though the exact algorithm behind this classification is not disclosed, our validation exercises (Appendix D) suggest that it is more selective than simple keyword matching and provides reliable coverage. This method is consistent with standard practice in the literature, including the construction of Economic Policy Uncertainty indices, see for instance Ghirelli et al. (2019) and Diakonova et al. (2026).

For each of the four economic regions (the U.S., the EU, Russia, and China) whose local GPRs we had previously computed, we select several focal regions of interest for examining the directed GPRs. Specifically, we focus on Middle East and North Africa (MENA, which for simplicity we abbreviate ME), Russia (or Russian Neighbours if the local country is Russia), China (or China Neighbours if the local country is China) – see Table F.1 for the exact constituents of each of these focal regions.

We now examine the resulting directed indices. They provide information beyond aggregate GPR in three complementary settings: identifying the geographical source of movements in aggregate risk, comparing the relative importance of different sources of perceived risk within a country, and comparing how different countries perceive risk associated with the same region.

**1. Comparing aggregate GPR with a specific directed component.** Aggregate GPR alone does not reveal which geographical source drives a particular increase in risk. Figure 3 illustrates this point for the United States and Russia. In the United States, the aggregate index rises considerably more after the 9/11 attacks than during the 2003 Iraq War. The U.S.–Middle East index, however, shows the opposite ranking: relative to its 2000 benchmark, Middle East-related GPR increases substantially more during the Iraq War than after 9/11. The geographical attribution of the aggregate movement is therefore not immediate from the aggregate series itself.

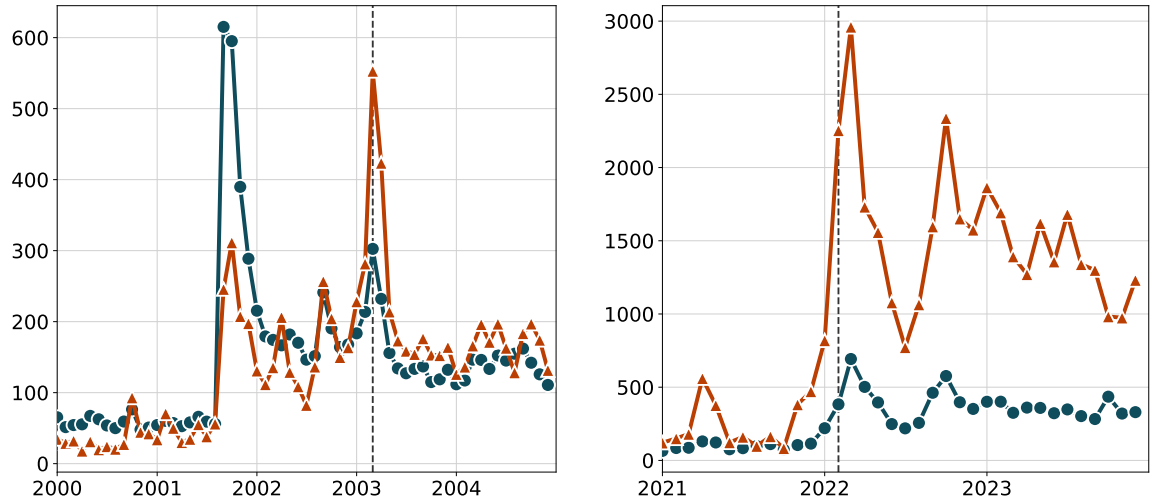
A similar distinction emerges for Russia. Around the full-scale invasion of Ukraine, aggregate Russian GPR rises by roughly a factor of five. By contrast, GPR directed towards Russia's neighbours begins increasing already in late 2021 and rises by roughly a factor of thirty. The directed index therefore captures both an earlier buildup and a substantially larger increase in the salience of risks associated specifically with neighbouring countries.

**2. Comparing a country's directed indices.** Directed indices also allow us to examine how the geographical composition of a country's geopolitical concerns changes over time. Figure 4 illustrates this for the United States by comparing *directed* GPR indices associated with the Middle East, Russia, and China. Each series displays pronounced increases around events involving its corresponding region, while their relative importance changes substantially over the sample.

Figure 3: GPR: General vs. directed

(a) US GPR (blue) and US GPR towards Middle East (orange)

(b) Russia's GPR (blue) and Russia's GPR towards its neighbours (orange)



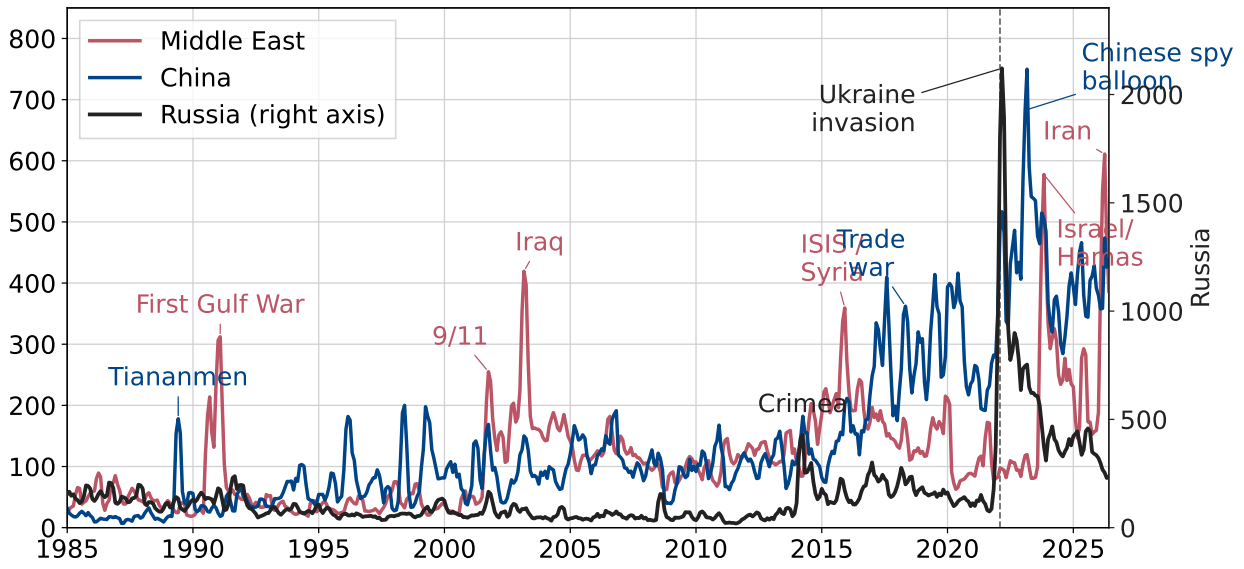

*Notes:* The figure compares aggregate country GPR (blue circles) with a directed GPR index (orange triangles). Subfigure 3a reports U.S. GPR directed towards the Middle East; Subfigure 3b reports Russian GPR directed towards Russia's neighbours. Indices are normalised to 100 in the year preceding the plotted sample. Russia's neighbours comprise Ukraine, Belarus, Georgia, and Armenia. See Table F.1 for the composition of the Middle East region.

The China-related index rises around the Tiananmen Square events and increases particularly strongly during the first Trump administration and the U.S.–China trade war.[12] Middle East-related GPR displays large increases around the major conflicts in the region. Russia-related GPR, meanwhile, rises by roughly a factor of five following the annexation of Crimea and by more than twenty following the full-scale invasion of Ukraine. These patterns illustrate how an aggregate GPR index can mask substantial shifts in the geographical composition of geopolitical concerns.

**3. Comparing different perspectives on the same source of risk.** Finally, directed indices allow us to compare how geopolitical risk associated with the same region is perceived across countries. Two dimensions are useful for this comparison. The first is the (raw) level of the index, which captures the relative prominence of a focal region in each country's geopolitical-risk coverage. The second is the change in attention around a particular episode, which we measure by normalizing each index to a common pre-event benchmark. The former captures differences in the underlying salience of a region, while the latter captures how strongly countries' attention shifts in response to a particu-

[12]Removing the term "*trade war*" from the query substantially smooths the index over precisely this period.

Figure 4: Comparing directed GPRs: how US perceives geopolitical risk associated with different regions




*Notes:* Directed U.S. GPR indices associated with the three focal regions shown in the legend. All indices are normalized to an average of 100 over 1985–2019. See Table F.1 for the exact composition of each region.

lar geopolitical event.

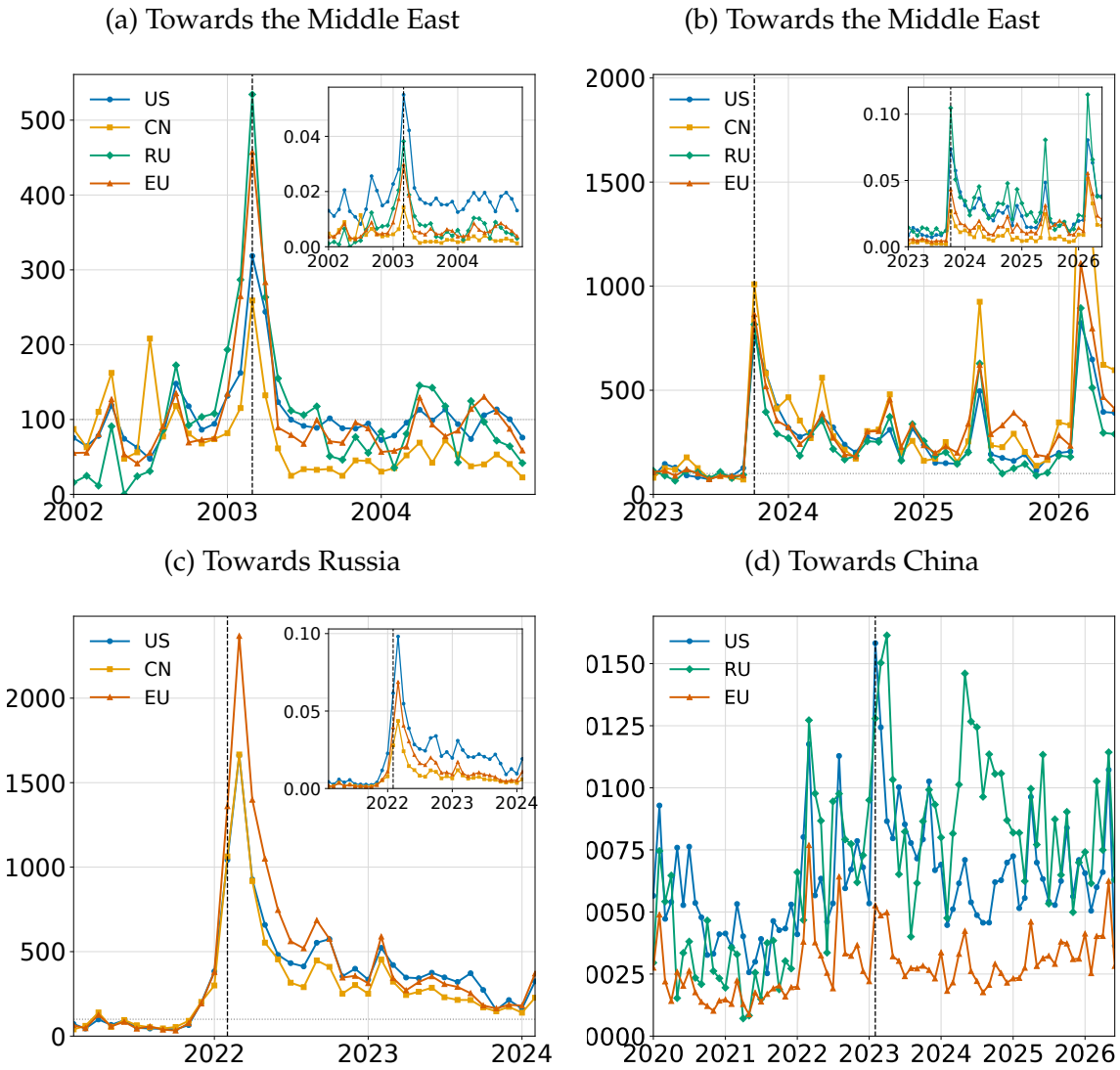
Figure 5 illustrates these two margins. During the 2003 Iraq War, Russia exhibits a substantially larger proportional increase in Middle East-related GPR than either the United States or China. In levels, however, U.S. coverage of Middle East-related geopolitical risk was already considerably higher before the war. Thus, a smaller proportional increase need not imply that the region is less important in the country's geopolitical narrative.

The more recent Middle East episodes provide a related example. From 2023 onward, the relative increases in Middle East-related GPR are initially similar across countries, but China's index rises particularly sharply around the escalation involving the Strait of Hormuz. Despite this larger proportional response, the level of Middle East-related GPR in China remains well below that observed in Russia and the United States.

Differences in perspective are also visible for Russia-related risk. Following the full-scale invasion of Ukraine, U.S. and Chinese attention to Russia increases by broadly similar proportions (subfigure 5c), although the U.S. starts from a higher level and subsequently remains more elevated. The increase in the EU is considerably larger, consistent with the much greater prominence of Russia in the European geopolitical-risk narrative during this episode.

The China-related indices (subfigure 5d) provide a final illustration. U.S. and Russian attention to China is persistently higher than that of the EU. The series also respond differ-

Figure 5: Directed attention to focal regions




*Notes:* Each subplot shows how that country (given by different lines) views geopolitical risk originating from that focal region (given by the subplot). The subplots towards Middle East and Russia are normalized to 100 in the preceding year; the inset are the raw fractions. Subfigure 5d shows just the raw fractions. The exact composition of the focal region can be found in Table F.1.

ently to subsequent events: U.S. attention rises sharply around the 2023 balloon episode, while Russian attention increases around Xi Jinping's visit to Moscow and remains elevated through 2024 as Russia–China relations deepen. These events are much less visible in the corresponding EU index. Taken together, these comparisons show that the same geopolitical source can differ substantially across countries both in its underlying salience and in the intensity with which particular events are perceived.

Now, while the directed framework can thus offer some clear advantages in under-

standing geopolitical risk, nevertheless, geopolitical narratives are often entangled. In terms of our methodology, that means that a single article may reference multiple regions, leading to overlapping directed indices. As a result, the sum of directed GPRs typically exceeds the total GPR index. While this overlap is informative for assessing relative importance, it prevents a clean decomposition into non-overlapping components. In the next section we address this challenge, describing how we use this framework to construct a non-overlapping decomposition of GPR by region, and present the resulting GPR profile for key countries.

# 4 Applications

In this section we present three empirical applications that illustrate the usefulness of our directed GPR indices. First, we decompose directed geopolitical risk across foreign regions to construct national GPR profiles and characterize the geographical composition of each country's geopolitical concerns. We then estimate the macroeconomic responses to GPR shocks originating from different regions and examine how their effects vary with the geographical source of risk. Finally, we conduct a "No-War" counterfactual to assess the macroeconomic effects of a major geopolitical episode and illustrate how geographically targeted GPR measures can complement aggregate indices.

## 4.1 National geopolitical risk profiles

Which regions account for a country's geopolitical risk, and how does their relative importance vary across countries? To address this question, we decompose each country's aggregate GPR into non-overlapping components associated with distinct geographical sources of risk. This decomposition allows us to construct country-specific geopolitical risk profiles that summarize the relative contribution of different regions to overall GPR.

GPR indices are based on the count of press articles related to geopolitical risk. When an article is focused on a specific region, it contributes to the corresponding directed GPR. Now, our first key observation is that a substantial share of GPR articles cannot be attributed to any external region. Such articles focus solely on domestic narratives or mention foreign regions too briefly to be captured by Factiva's regional classification system. They often reference geopolitical events only in passing or focus on purely domestic consequences. Appendix E provides details on how our methodology is able to identify such components, as well as a detailed narrative taxonomy of GPR-relevant articles that are not focused on any foreign region. As shown in Figure E.1 in the Appendix, such *internal*

component share can be non-trivial — the monthly maximums can reach 79% for Russia and 65% for the U.S. What is clear is that in all four regions the second decade of the decomposition sample, 2013–2023, saw a significant decline in this self-centered (*internal*) share of GPR: the mean fell from 44% to 32% in the U.S., 42% to 35% in the EU, 61% to 37% in Russia, and 51% to 25% in China, with differences that are statistically significant.

Because our objective is to identify the geographical sources of outward-looking risk, we set aside this *internal* component and decompose the remaining *external* part. This distinction concerns the geographical focus of news coverage: articles classified as *internal* may still reflect the domestic repercussions of events originating abroad. Future research could examine this component in greater detail, distinguishing incidental references to geopolitical events from substantive coverage of their domestic consequences, and assessing whether it provides additional information about the transmission of geopolitical risk to economic activity.

Now, our main challenge is a methodological one, that of overlapping coverage. Consider a simplified case of a country $C$ with two external regions, $A$ and $B$. Articles in the press of $C$ that mention both regions would contribute to both directed indices, inflating their sum relative to the external GPR. To resolve this, we construct a set of queries designed to compute *all possible region combinations*. For example, we separately count the number of articles that mention only $A$ and not $B$, only $B$ and not $A$, and both $A$ and $B$. This approach ensures that each article contributes to one and only one directed component. We label indices constructed using such mutually exclusive combinations *exclusive* directed indices[13]. When such numerators are divided by the total number of articles contributing to the external part of the GPR, we obtain a decomposition of external risk into non-overlapping components. In other words, the total count of all the articles associated with all the various components, added to those that contribute to the *internal* part, make up the numerator in the standard GPR index.

This method introduces two practical challenges. First, the number of region combinations grows exponentially with the number of regions ($2^n$ for $n$ regions). Second, Factiva imposes strict limits on query length, which constrains the number of regions we can include in a single query[14]. To address both issues, we group regions into broad geopolitical groups. This allows us to maintain interpretability while ensuring sufficient coverage and comparability across countries[15].

[13] By contrast, we sometimes label the default directed indices that rely only on an explicit inclusion of a region as "*marginal*".

[14] As of 2026 Factiva amplified this maximal query length two-fold.

[15] The "*re*" filter is particularly useful here, as it supports nested classifications and flexible region definitions.

For each country whose geopolitical risk profile we want to compute, we define a small set of relevant external regions based on geopolitical salience and media coverage. One aim of having broad regions is to decrease the possibility of interpreting noise as signal, as articles very with a very precise geographical focus are likely to be much rarer. Such regions of interest typically include the Middle East and North Africa (MENA), a variant of the "Western bloc", and either China, Russia, or their neighbors. Table F.1 shows the regional breakdown for the U.S., China, Russia, and the EU; Appendix F.1 details the validation of these regional groupings.

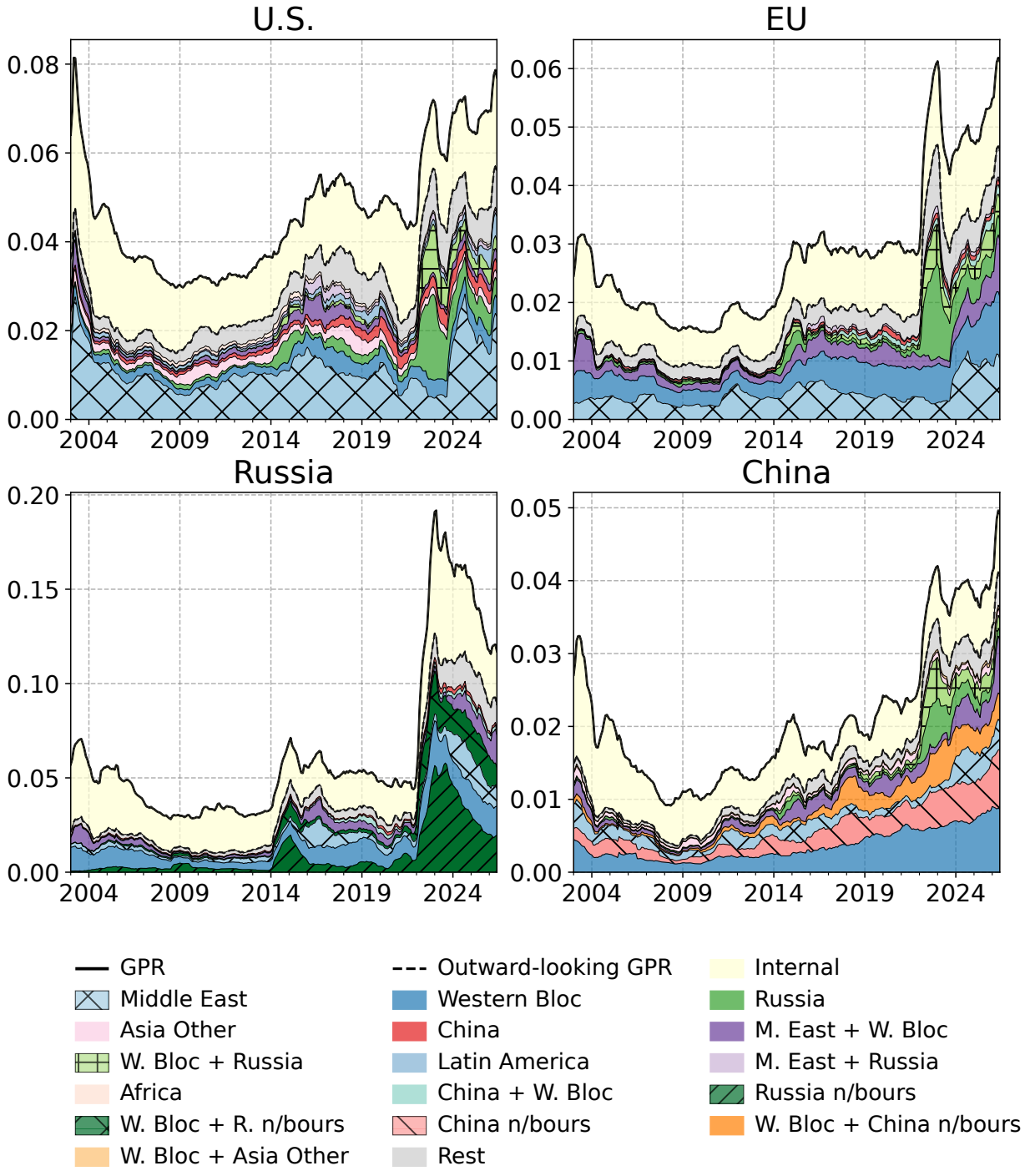
Our procedure is as follows. For each country (e.g. the U.S.), we compute the number of articles associated with every possible combination of the defined regions (e.g., $2^5 = 32$ combinations for five regions). We then rank these combinations by article count and retain the top $N$ combinations that together account for at least 80% of the external GPR. These selected combinations form the basis for our exclusive, high-frequency directed GPR indices. This approach avoids double-counting and enables a clearer understanding of how different regions contribute to a country's geopolitical risk narrative.

The resulting geopolitical risk profiles for the U.S., the EU, China, and Russia are shown in Figure 6.[16] The profiles reveal substantial heterogeneity in both the geographical composition of risk across countries and its evolution over time. The U.S. and the EU display broadly similar patterns, with the Middle East and North Africa (MENA) representing a major source of geopolitical risk, although attention shifts toward Russia and, to a lesser extent, China in more recent years. In the U.S., the exclusively MENA-related component accounts for almost 50% of external GPR coverage in the 2000s, but roughly half that share following Russia's full-scale invasion of Ukraine. In the EU, the composition is more balanced: MENA and the Western Bloc, predominantly the U.S., account for large shares of external GPR, while Russia rises to comparable prominence following the invasion.

These patterns contrast with China's persistent focus on the Western Bloc and Russia's increasing attention to its immediate neighbours. The profiles also reveal substantial asymmetries in bilateral geopolitical concerns. Russia, for example, becomes considerably more prominent in China's GPR narrative following the invasion of Ukraine, whereas China remains a relatively small component of Russia's external GPR. More generally, the decomposition captures both gradual shifts in the geographical composition of geopolitical risk and sharp reallocations of attention around major geopolitical events. It therefore provides a compact characterization of which regions matter for each country's

[16] Figure G.1 in the Appendix presents an alternative normalization and reports the average contribution of each component to external GPR.

Figure 6: Country Geopolitical Risk Profiles




*Notes:* Each panel shows the main components of the country's GPR decomposition. Each component represents the share of GPR articles focused on the indicated region or combination, excluding all other regions listed for that country. For example, "Middle East + Western Bloc" includes articles focused on both regions but none of the other specified regions. Regional definitions vary across countries, as detailed in Table F.1; as well as the specific regions focused on. The *internal* component does not focus on any foreign region(s). Data are monthly values (number of articles divided by the total number of articles) smoothed with a one-year rolling window. We start with 2003 due to the first availability of the decomposition for the U.S.

Table 2: Exclusive versus joint regional coverage in European external GPR, 2020–2026.

| Region | Solo (%) | Joint (%) | Total involving (%) |
|---|---|---|---|
| Russia | 15 | 14 | 29 |
| Middle East | 19 | 16 | 35 |
| Western Bloc | 21 | 27 | 48 |
| China | 2 | 7 | 9 |

*Notes:* Solo is the exclusive share of articles about that region alone; joint sums all exclusive combinations that include the region together with at least one other focal region. The numbers are percentages of outward-looking GPR.

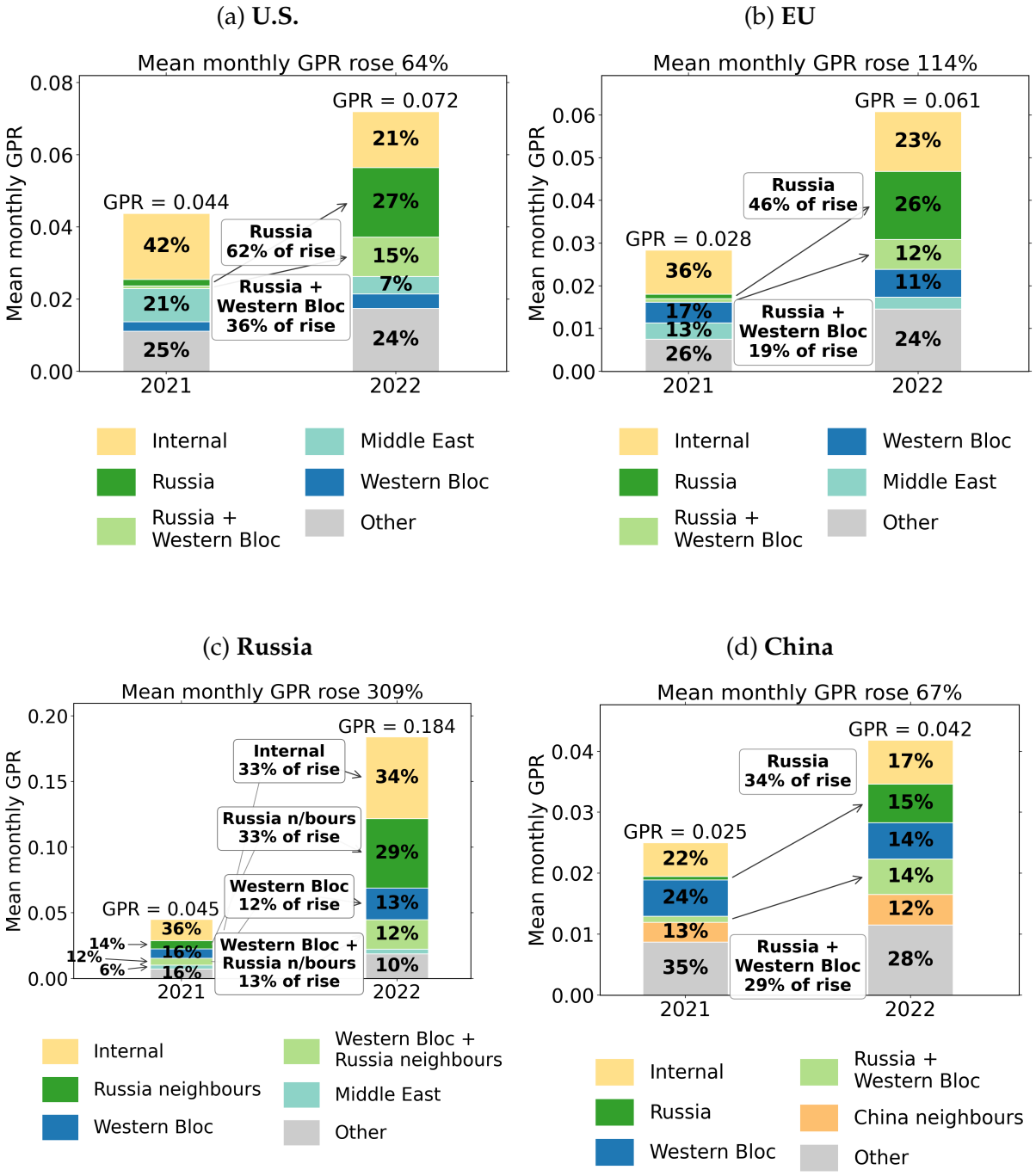
geopolitical risk narrative and how these profiles evolve over time.

A second notable point is that coverage involving multiple regions forms a substantial part of geopolitical risk, making explicit treatment of overlap essential. In each of the four economic regions considered, a multi-region combination ranks among the six largest components. Table 2 shows why that distinction matters on an example of the EU GPR decomposition. For Russia and the Middle East, exclusive coverage still exceeds joint coverage: each is often discussed on its own. For the Western Bloc, and especially for China, the opposite holds—China's solo share is only about 2% of European external GPR, but a further 7% comes from multi-region combinations. These joint narratives add information about the relationships through which geopolitical concerns are framed: Chinese coverage linking Russia and the Western Bloc emphasizes sanctions and strategic confrontation, while coverage linking the Western Bloc and China's neighbors centers on regional security tensions. Our decomposition preserves these connections without counting the same article more than once. Together, the results show why the level of aggregate GPR alone cannot reveal either the changing geographical focus of risk or the combinations of actors involved.

We next illustrate the usefulness of the non-overlapping decomposition in two episodes: Russia's full-scale invasion of Ukraine in 2022 and the escalation of conflict in the Middle East beginning in October 2023. The two episodes differ in an important respect. The invasion of Ukraine was accompanied by a large increase in aggregate GPR and in the components driving it. By contrast, the post-October 2023 episode involved a much smaller change in aggregate GPR, but a substantial reallocation of attention across geographical components. These are precisely the margins along which the decomposition is most informative: it identifies which regions account for changes in aggregate GPR, how their contributions evolve, and how the underlying geopolitical narrative is reconfigured around major events.

The 2022 episode is shown in Figure 7. Average monthly aggregate GPR increased

Figure 7: Change in GPR composition in 2022




*Notes:* Each panel shows the main components of the country's GPR decomposition. Each component represents the share of GPR articles focused on the indicated region or combination, excluding all other regions listed for that country in Table F.1. For example, "Middle East + Western Bloc" includes articles focused on both regions but none of the other specified regions. The *internal* component does not focus on any foreign region(s). The callouts report the percentage of the rise (note that this is not the same as the change in percentage points).

sharply in all four economies, ranging from around 60% in the U.S. and more than 100% in Europe to over 300% in Russia. In the U.S., virtually the entire increase is associated with Russia and narratives jointly involving Russia and the Western Bloc. In Europe, the increase is more broadly distributed across coverage focused on Russia, the Western Bloc, and Europe itself. In particular, the *internal* component rises by around 11%, consistent with greater attention to the domestic consequences of the invasion, alongside smaller increases in components jointly involving Russia and other regions.[17] China displays a different pattern. The increase in coverage jointly involving Russia and the Western Bloc is comparable to that of Russia-only coverage, suggesting that Chinese media placed substantial emphasis on the broader geopolitical implications of the war. Finally, in Russia, the increase is concentrated primarily in the *internal* component and coverage associated with neighboring countries, particularly Ukraine, with a secondary contribution from narratives involving the Western Bloc.

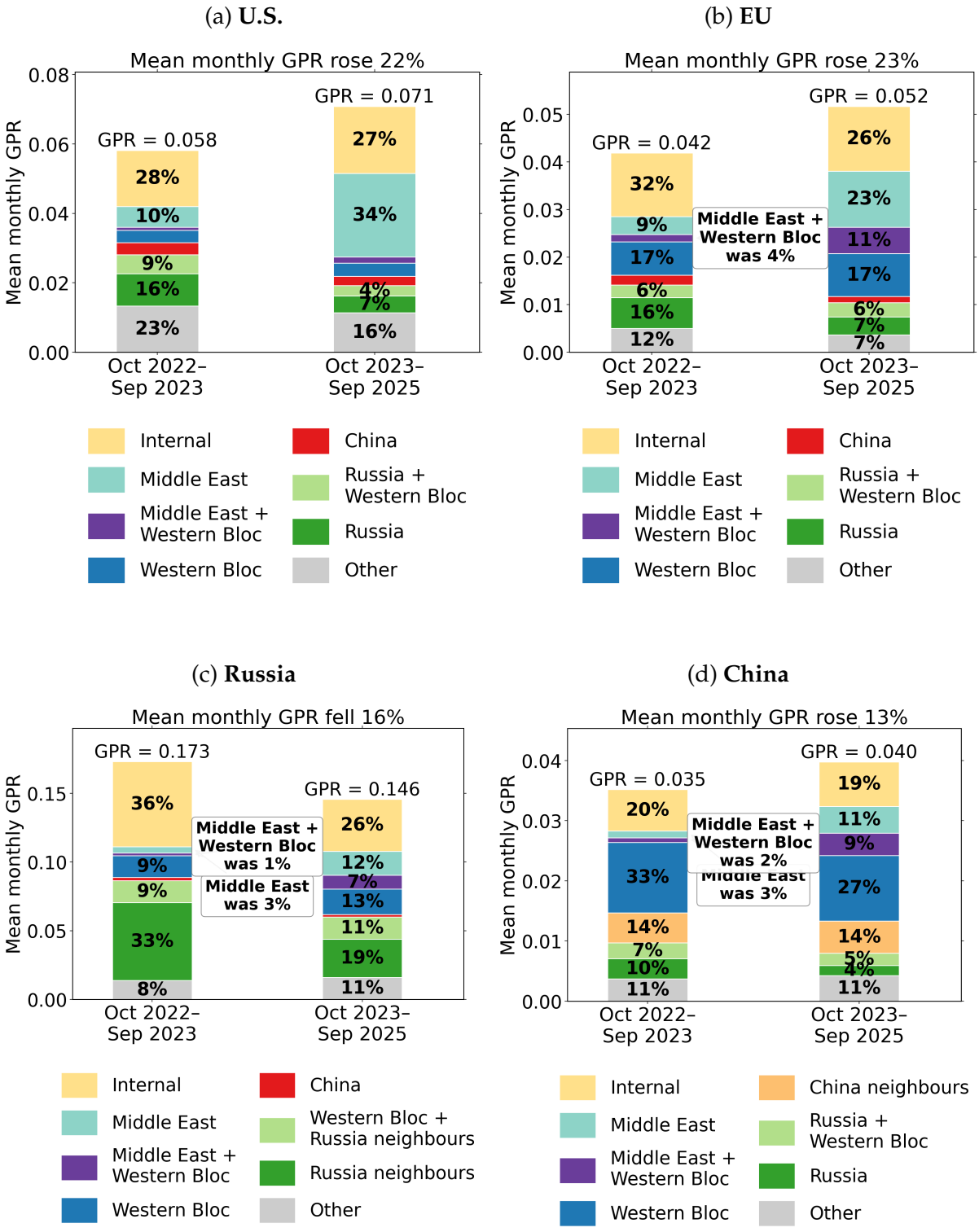
The episode beginning in October 2023 looks quite different. Aggregate GPR rises only modestly in the U.S. and Europe, by around 20%, and even less in China, by 13%, while declining by 16% in Russia. Yet the geographical composition of GPR changes substantially, as shown in Figure 8. The share of coverage focused exclusively on the Middle East increases by 24 percentage points in the U.S., 14 percentage points in the EU, 11 percentage points in Russia, and 7 percentage points in China. Coverage jointly involving the Middle East and the Western Bloc also rises by around 7 percentage points in the EU, Russia, and China, but not in the U.S. One possible explanation is that U.S. involvement in the episode is naturally part of the domestic narrative and therefore does not appear as a separate foreign-region component in the U.S. decomposition.[18]

This reallocation toward the Middle East is accompanied by a decline in Russia-related coverage outside Russia, which roughly halves in several cases. Europe is a partial exception: while attention to Russia- and Ukraine-specific developments declines, the component jointly involving Russia and the Western Bloc remains broadly unchanged, suggesting continued attention to the wider implications of the war for Europe and its allies. A similar distinction appears in Russian media. Internal coverage and reporting focused on Ukraine decline, while the component linking Russia's neighboring countries to the Western Bloc is the only Russia-related component that remains stable or increases. The episode therefore illustrates how substantial changes in the geographical composition of

[17] For the U.S., the Western Bloc consists predominantly of Europe; for the EU, it includes the U.S. and other Western countries.

[18] Accordingly, U.S. articles discussing the Middle East together with U.S. involvement can still be classified as Middle East-only from the perspective of the geographical decomposition, consistent with the larger increase in the U.S. Middle East component.

Figure 8: Change in GPR composition after October 2023




*Notes:* Each panel shows the main components of the country's GPR decomposition. Each component represents the share of GPR articles focused on the indicated region or combination, excluding all other regions listed for that country in Table F.1. For example, "Middle East + Western Bloc" includes articles focused on both regions but none of the other specified regions. The *internal* component does not focus on any foreign region(s). The callouts report the percentage of the GPRs not visible in the main plots.

geopolitical risk can occur even when the aggregate GPR index moves relatively little.

## 4.2 Revisiting the macroeconomic effects of geopolitical risk shocks

We illustrate the empirical value of our directed indices by revisiting the macroeconomic effects of geopolitical risk. Using Bayesian structural VARs, we examine how economic activity, inflation, and monetary policy respond to innovations in aggregate GPR and its *directed* components. We incorporate the COVID-19 volatility adjustment of Lenza and Primiceri (2022), allowing us to combine a long pre-pandemic sample with recent geopolitical events within a consistent framework. This approach allows us to assess how the geographical source of risk shapes the timing, persistence, and magnitude of the macroeconomic effects. We subsequently use the estimated models to construct a counterfactual assessing the effects of the increase in geopolitical risk associated with the onset of the conflic in the Middle East in early 2026 and its implications for the U.S. economy.

### 4.2.1 Data

In this section, we focus on monthly data for China, Russia and the United States, with all samples ending in June 2026. The Russian sample begins in January 2002, the Chinese sample in January 1995, while the U.S. sample starts in January 1986. The differences arise from the historical availability of the geopolitical risk indices. Each VAR includes six variables: the directed geopolitical risk index of interest, real GDP, consumer prices, a measure of the monetary policy stance, equity-market volatility, and real commodity prices. We estimate separate specifications using the aggregate GPR index and the selected directed indices to compare the macroeconomic effects of geopolitical risk originating in different locations.

For the three economies, inflation is measured using the consumer price index. We proxy the monetary policy stance with the policy interest rate for Russia, the one-year benchmark deposit rate set by the People's Bank of China from the China Macroeconomy Time Series database provided by the Federal Reserve Bank of Atlanta for China (Chang et al., 2016; Higgins and Zha, 2026), and the one-year Treasury yield for the United States. Equity-market volatility measures financial uncertainty. For China and Russia, we construct realized volatility as the sum of squared equity returns; for the United States, we use the CBOE Volatility Index (VIX). Commodity prices capture an international transmission channel of particular relevance to the three economies. We use Urals crude oil prices for Russia, copper prices for China, and WTI crude oil prices for the United States.

Series are seasonally adjusted where appropriate, and commodity prices are expressed in real terms. Variables available only at quarterly frequency are converted to monthly frequency using cubic-spline interpolation. Further details on variable definitions, transformations, and data sources are provided in Appendix H.

#### 4.2.2 Methodology: Bayesian SVAR with COVID-19 volatility correction

We estimate a Bayesian SVAR augmented with the COVID-19 volatility adjustment proposed by Lenza and Primiceri (2022). This specification allows us to retain the full estimation sample while limiting the influence of the exceptionally large variance of the macroeconomic shocks associated with the outbreak of the pandemic.

For each country, the reduced-form VAR is specified as

$$y_t = C + \sum_{\ell=1}^{p} B_\ell y_{t-\ell} + s_t \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \Sigma), \tag{1}$$

where $y_t$ is an $n \times 1$ vector containing the geopolitical risk indicator and the macroeconomic, $C$ is an $n \times 1$ vector of constants, $B_\ell$ is an $n \times n$ matrix of autoregressive coefficients associated with lag $\ell$, and $\varepsilon_t$ is an $n \times 1$ vector of reduced-form innovations. The factor $s_t$ rescales the residual covariance matrix during the pandemic period. More precisely, let $t^*$ denote the onset of the pandemic in February 2020. The scaling factor is equal to 1 before $t^*$, so observations prior to the pandemic are governed by the baseline covariance matrix $\Sigma$. Following Lenza and Primiceri (2022), we allow the scaling factor to take separate values during the first three months of the pandemic:

$$s_{t^*} = \bar{s}_0, \qquad s_{t^*+1} = \bar{s}_1, \qquad s_{t^*+2} = \bar{s}_2. \tag{2}$$

Thereafter, the scaling factor is assumed to converge gradually toward its pre-pandemic value according to

$$s_{t^*+j} = 1 + (\bar{s}_2 - 1)\,\rho^{j-2}, \qquad j \geq 3, \tag{3}$$

For estimation, we rely on a conjugate Normal–Inverse-Wishart prior for the VAR coefficients and covariance matrix. In particular, we adopt a standard Minnesota prior (Litterman, 1986), with tightness chosen following Giannone et al. (2015). For $\bar{s}_0$, $\bar{s}_1$, and $\bar{s}_2$, we use Pareto priors with scale and shape parameters equal to one, which have a very fat right tail, and are thus consistent with possible large increases in the variance of the VAR innovations. For $\rho$, we use a Beta prior with mode 0.8 and standard deviation 0.2, following Lenza and Primiceri (2022). The VAR includes 12 monthly lags, and impulse

responses are computed over a 24-month horizon. We report posterior medians together with 68% and 90% credible bands. Posterior inference is based on 5,000 draws, with the first 2,500 discarded as burn-in.

We identify geopolitical risk shocks recursively, following Caldara and Iacoviello (2022) and Bondarenko et al. (2024). In each country VAR, the relevant GPR index is ordered first in the Cholesky decomposition of the reduced-form covariance matrix. Under this ordering, a GPR shock can affect all other variables contemporaneously, while shocks to the remaining variables affect GPR only with a lag.

#### 4.2.3 Empirical results

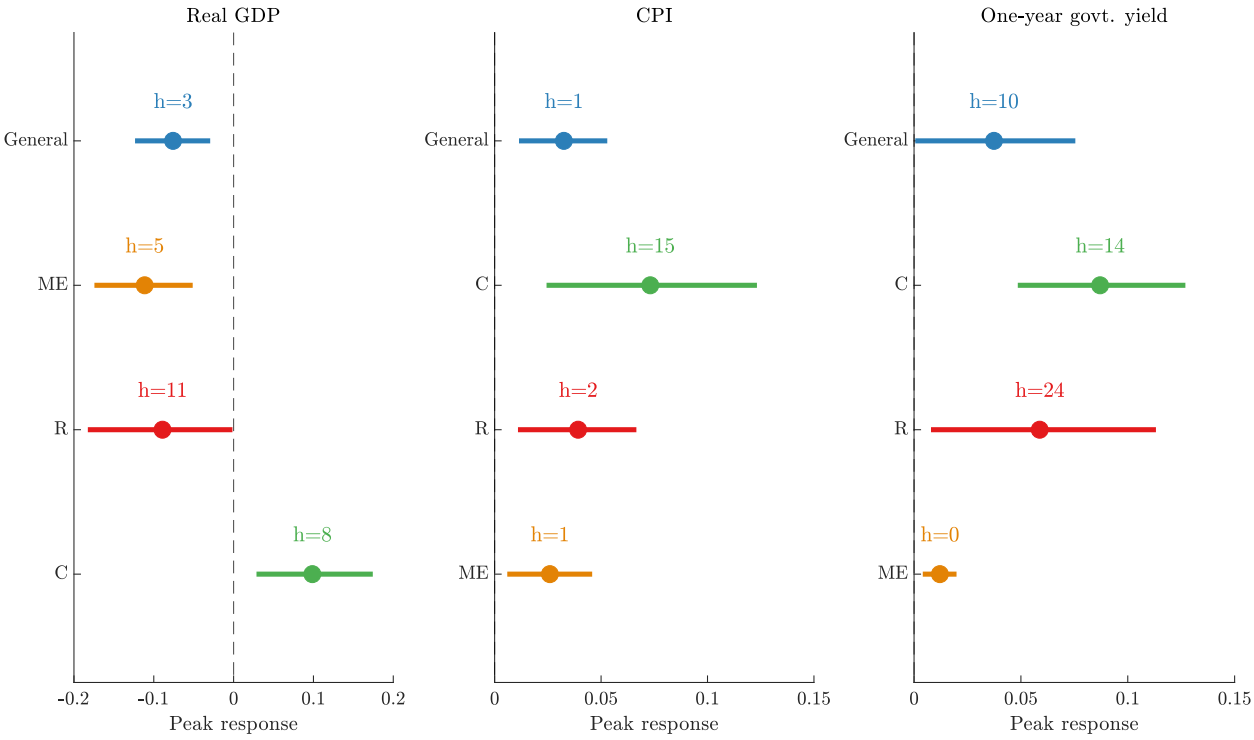
We begin by examining how the U.S., Chinese and Russian economies respond to geopolitical risk originating in different foreign regions. Figures 9, 10 and 11 summarize the magnitude and timing of the peak responses to one-standard-deviation shocks to the aggregate and directed GPR indices[19]. The complete set of impulse responses is shown in Appendix H.

Aggregate GPR shocks are followed in the three economies by lower economic activity, higher prices, and a tightening of monetary policy. This pattern is consistent with an adverse supply-type transmission mechanism and with the evidence in Bondarenko et al. (2024, 2025) and Caldara et al. (2026). The directed indices, however, reveal substantial heterogeneity that is masked by aggregate measures of geopolitical risk. In particular, the magnitude, timing, and, in some cases, the sign of the macroeconomic responses depend on the geographical source of the shock.

In the United States, the aggregate GPR shock leads to a decline in real GDP of approximately 0.08 percentage points at the peak, with the trough occurring after three months. Middle East and Russia-related shocks lead to somewhat more delayed contractions, of approximately 0.11 and 0.08 percentage points, respectively, with both troughs occurring after five months. Instead, China-originating shocks trigger a markedly different response: GDP increases, reaching a peak of approximately 0.10 percentage points after eight months. Consumer prices rise in all specifications, with the largest response associated with China-related shocks, which also bring about the strongest increase in the one-year Treasury yield, of approximately 0.09 percent. The responses to Middle East and Russia-related shocks therefore resemble those of an adverse supply disturbance, with activity declining as prices and interest rates rise. China-related shocks, by contrast, are followed by higher output alongside higher prices and interest rates, pointing to a dif-

[19] In particular, we report the largest absolute posterior median among 68% significant horizons (credible interval excludes zero); if none are significant, we report the largest absolute posterior median.

Figure 9: Peak macroeconomic responses to geopolitical risk shocks in the U.S.

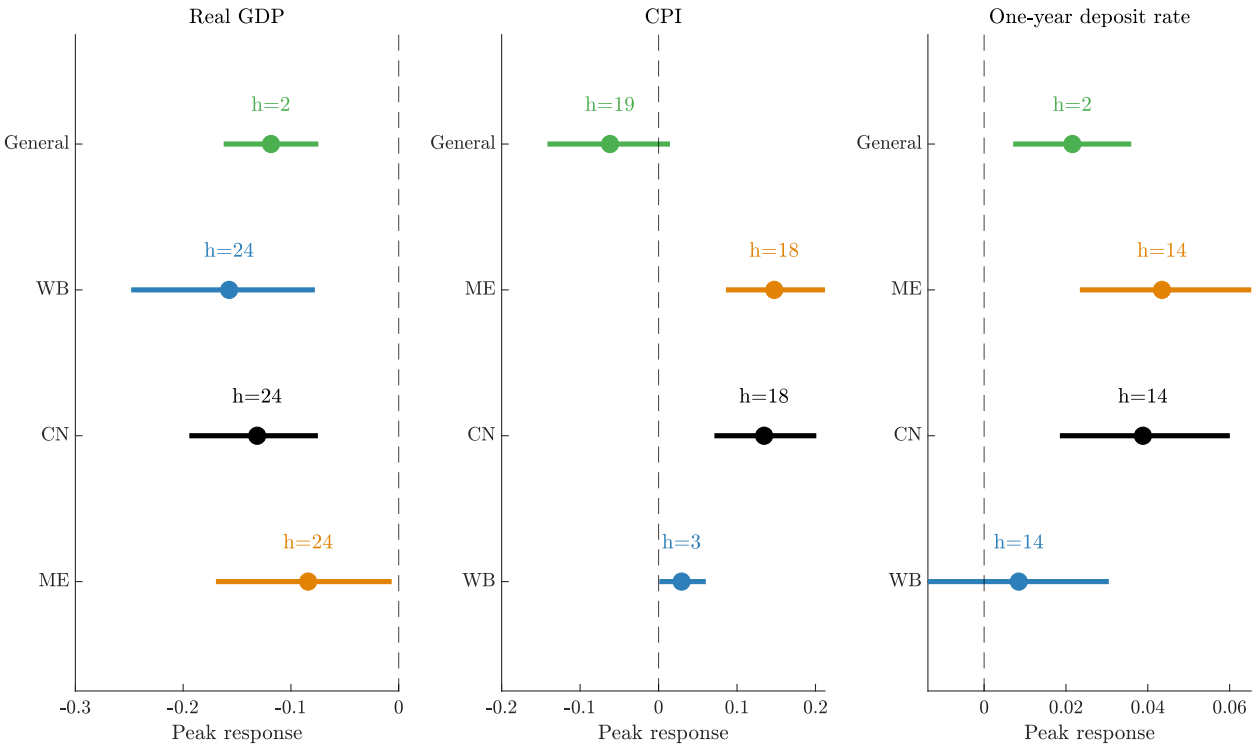



*Notes:* The figure reports the peak responses of U.S. real GDP, consumer prices, and the two-year Treasury yield to one-standard-deviation shocks to the aggregate GPR index and directed GPR indices associated with the Middle East (ME), Russia (R), and China (C). Only peak responses whose 68% credible intervals exclude zero are reported. These regions are selected based on their prominence in recent U.S. geopolitical risk coverage (Figure 6 and the numbers behind it). Responses are estimated using Bayesian structural VARs over 1986M01–2026M06, each including the relevant GPR index, real GDP, the consumer price index, the one-year Treasury yield, the CBOE Volatility Index (VIX), and the real WTI oil price. Real GDP and consumer-price responses are expressed in percentage points, and Treasury yield responses in percent. Dots denote posterior median estimates of the peak responses, and horizontal bars denote 68% credible intervals. Labels ($h$) indicate the number of months after the shock at which the peak occurs. "General" denotes the specification using the aggregate GPR index.

ferent pattern of transmission. This latter result echoes the cross-country heterogeneity documented by Caldara et al. (2026), who identify the United States as a notable outlier: unlike most countries in their sample, U.S. GDP rises following a country-specific GPR shock, a pattern they associate with the country's distinct geopolitical position and more limited exposure to direct supply-side disruptions. Our directed decomposition shows that this expansionary response is not uniform across geographical sources of risk.

In both China and Russia, GPR shocks are recessionary, but their magnitude, timing, and price effects depend markedly on their geographical origin. In Russia, the aggregate shock produces the largest GDP decline, approximately 0.33 percentage points after seven months, while bilateral shocks generate somewhat smaller contractions. Middle East and

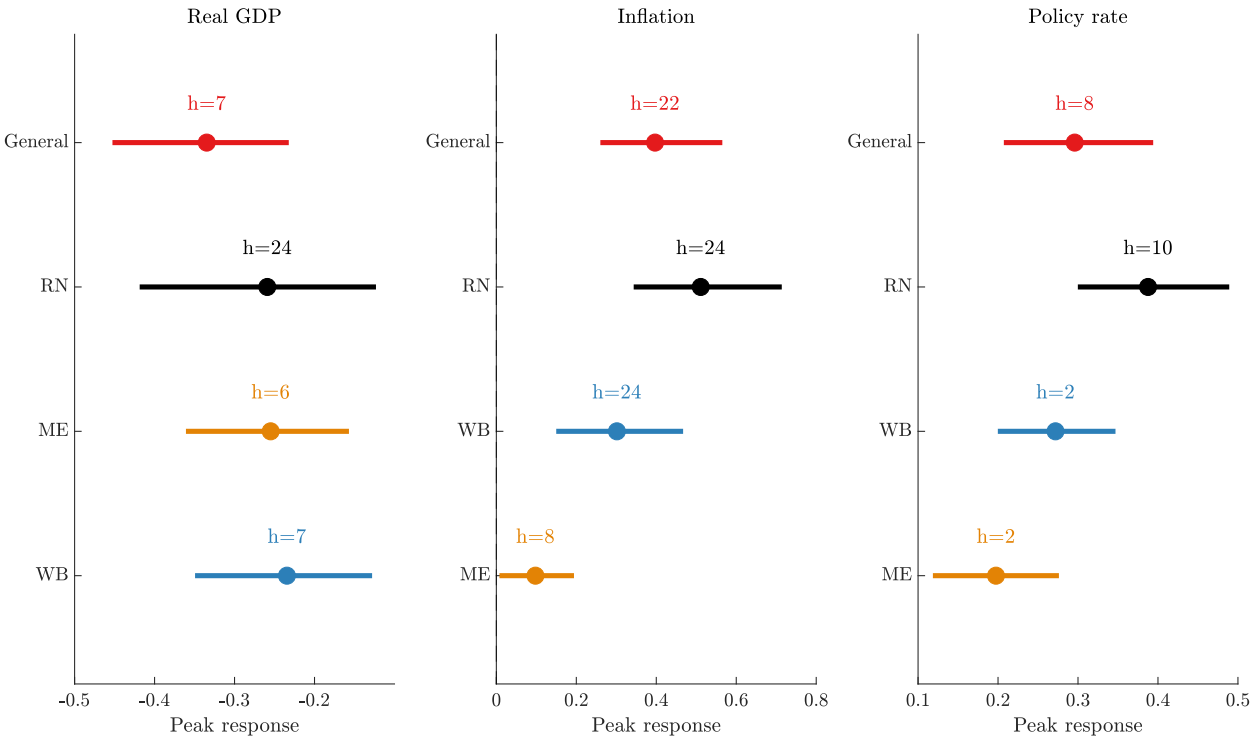
Figure 10: Peak macroeconomic responses to geopolitical risk shocks in China




*Notes:* The figure reports the peak responses of Chinese real GDP, consumer prices, and the one-year benchmark deposit rate to one-standard-deviation shocks to the aggregate GPR index and directed GPR indices associated with the Middle East (ME), the Western bloc (WB), and China's neighbors (CN). These regions are selected based on their prominence in recent Chinese geopolitical risk coverage (Figure 6 and the numbers behind it). Responses are estimated using Bayesian structural VARs over 1995M01–2026M06, each including the relevant GPR index, real GDP, the consumer price index, the one-year benchmark deposit rate, a measure of financial-market volatility, and real copper prices. Real GDP and consumer-price responses are expressed in percentage points, and deposit rate responses in percent. Dots denote posterior median estimates of the peak responses, and horizontal bars denote 68% credible intervals. Labels ($h$) indicate the number of months after the shock at which the peak occurs. "General" denotes the specification using the aggregate GPR index.

Western Bloc shocks affect activity relatively quickly, whereas the contraction following risk originating in Russia's neighbors develops gradually and reaches its trough only after almost two years. In China, the aggregate shock also produces a rapid contraction, with GDP reaching its trough after two months. By contrast, shocks associated with the Western Bloc, China's neighbors, and the Middle East have substantially more delayed effects, with output continuing to decline through the end of the two-year horizon; among these, the Western Bloc shock produces the largest contraction. The price responses differ more sharply across the two economies. In Russia, all GPR shocks raise consumer prices and elicit monetary tightening, although the timing and magnitude of these responses vary considerably. In China, the aggregate shock lowers consumer prices, although not

Figure 11: Peak macroeconomic responses to geopolitical risk shocks in Russia




*Notes:* The figure reports the peak responses of Russian real GDP, consumer prices, and the policy rate to one-standard-deviation shocks to the aggregate GPR index and directed GPR indices associated with the Middle East (ME), Russia's neighbors (RN), and the Western Bloc (WB). These regions are selected based on their prominence in recent Russian geopolitical risk coverage (Figure 6 and the numbers behind it). Responses are estimated using Bayesian structural VARs over 2002M01–2026M06, each including the relevant GPR index, real GDP, the consumer price index, the policy rate, a measure of financial-market volatility, and the real Urals oil price. Real GDP and consumer-price responses are expressed in percentage points, and policy-rate responses in percent. Dots denote posterior median estimates of the peak responses, and horizontal bars denote 68% credible intervals. Labels ($h$) indicate the number of months after the shock at which the peak occurs. "General" denotes the specification using the aggregate GPR index.

statistically significant, whereas the three bilateral shocks raise them. The Russian results are broadly consistent with Bondarenko et al. (2024), who document significant recessionary and inflationary effects of locally measured GPR shocks. Our bilateral decomposition extends this evidence by showing that the transmission of geopolitical risk varies considerably with its geographical source, particularly in the timing, persistence and even sign of the responses in some cases.

## 4.3 "No-War" counterfactuals for the U.S.

With the estimates from the previous section at hand, we next assess the macroeconomic importance of the sharp increase in geopolitical risk surrounding the escalation of the

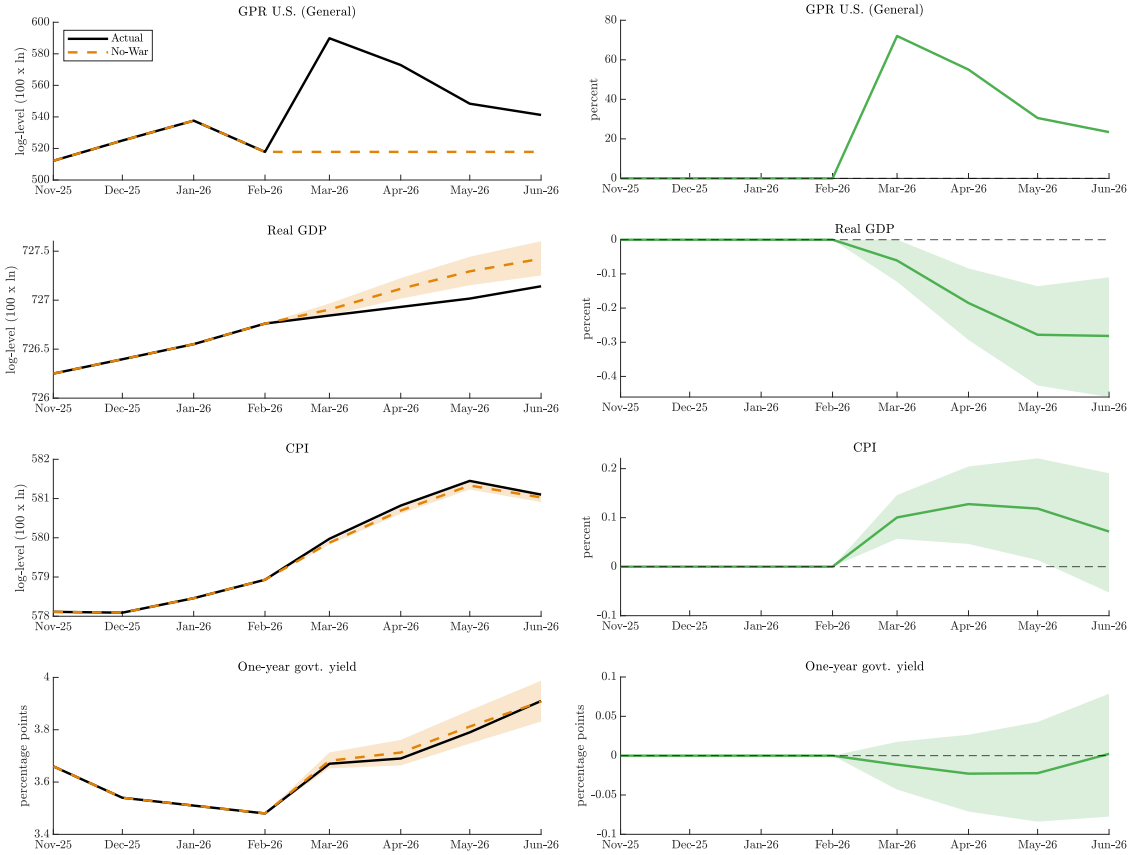
Middle East conflict in late February 2026, following the onset of U.S. and Israeli military action against Iran. To this end, we construct "No-War" counterfactuals that trace the model-implied path of the U.S. economy had geopolitical risk remained at its February 2026 level. Related counterfactual exercises have been used by Caldara et al. (2026) to quantify the global effects of the rise in geopolitical risk surrounding Russia's full-scale invasion of Ukraine and by Bondarenko et al. (2025) to assess its consequences for the EU. Our setting allows us to extend this approach by exploiting the geographical information contained in our directed indices. Since the source of the geopolitical tensions is known, we can compare a counterfactual based on aggregate U.S. GPR with one based specifically on U.S.–Middle East GPR. This provides a direct application in which our directed measures can complement existing aggregate GPR indices.

We first compute a counterfactual in which the aggregate U.S. GPR index is held fixed at its February 2026 level through June 2026, thereby removing the subsequent increase in geopolitical risk associated with the conflict. Operationally, we closely follow Bondarenko et al. (2025) by imposing a flat path for the GPR index and allowing the identified structural GPR shock to adjust endogenously so that the VAR reproduces this trajectory (see, e.g., Waggoner and Zha, 1999; Sims and Zha, 2006; Hubrich et al., 2026). All other structural innovations are left unchanged. The resulting counterfactual therefore isolates the contribution of the increase in geopolitical risk captured by the identified GPR shock, holding fixed all other structural innovations implied by the VAR.

Figure 12 reports the results. The left-hand panels compare the observed variables with their "No-War" paths. Given the Bayesian estimation framework, the counterfactual yields a posterior distribution for each macroeconomic path; orange dashed lines denote posterior medians and orange shaded areas the corresponding 68% credible intervals. The right-hand panels report the actual-minus-counterfactual gaps, with green solid lines denoting posterior medians and green shaded areas the corresponding 68% credible intervals.

Relative to the "No-War" counterfactual, the aggregate U.S. GPR index rises by approximately 70% in March 2026, before gradually declining over the following months. The associated macroeconomic effects build more gradually. Real GDP falls progressively below its counterfactual path, with the gap reaching approximately 0.3% by June. Consumer prices rise by around 0.1% relative to the counterfactual during the spring, although the difference subsequently narrows. By contrast, the response of the one-year Treasury yield is small and imprecisely estimated, with its median path lying slightly below the "No-War" counterpart during April and May. Overall, the aggregate-GPR counterfactual points to weaker U.S. activity together with temporary upward pressure on

Figure 12: Iran conflict: "No-War" counterfactual using aggregate U.S. GPR



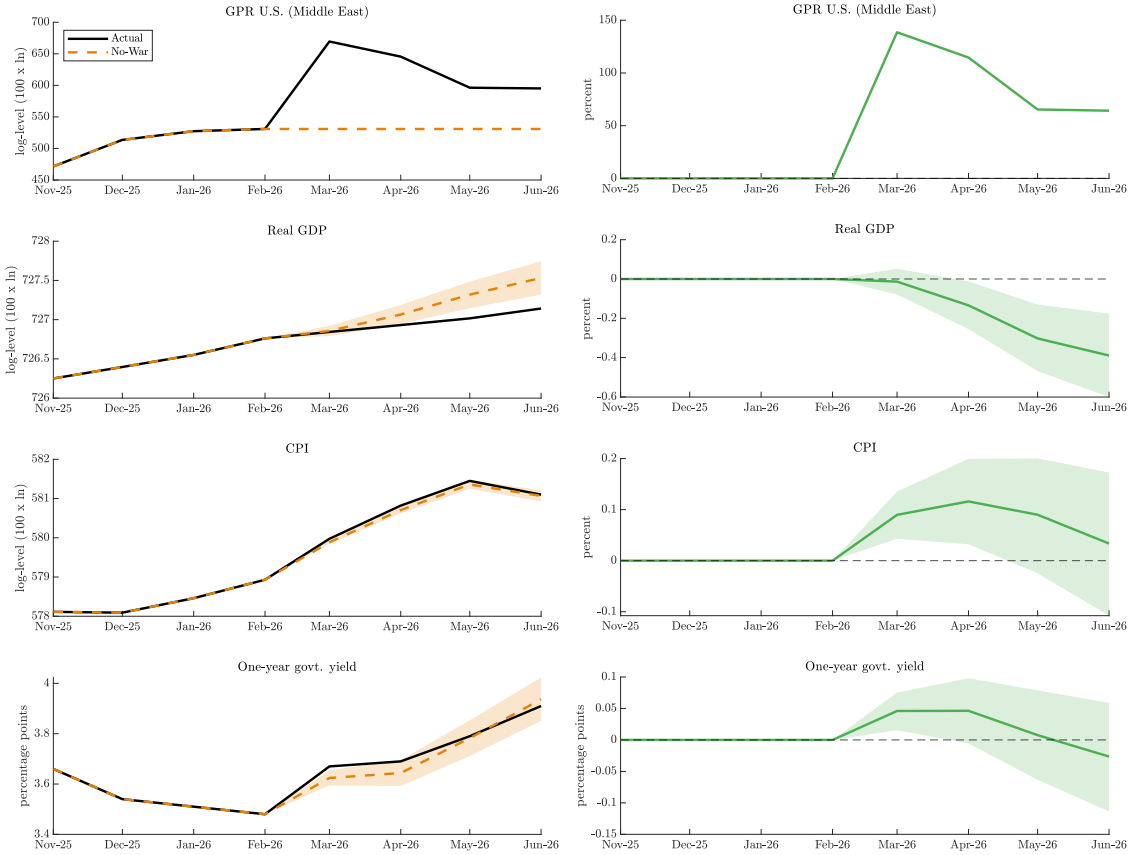

*Notes:* The left-hand panels report the observed paths of the aggregate U.S. GPR index, real GDP, consumer prices, and the one-year Treasury yield (solid lines), together with their "No-War" counterfactual paths (orange dashed lines). The counterfactual holds the aggregate U.S. GPR index fixed at its February 2026 level through June 2026 by allowing the identified structural GPR shock to adjust endogenously. Orange shaded areas denote 68% credible intervals. The right-hand panels report the difference between the actual and counterfactual paths. Green lines denote posterior medians and shaded areas the corresponding 68% credible intervals. Real GDP and consumer-price differences are expressed in percent, and Treasury-yield differences in percentage points.

consumer prices.

The aggregate index, however, combines geopolitical developments originating in different parts of the world. The Iran conflict therefore provides a natural setting in which to exploit the geographical information contained in our directed indices. We repeat the exercise using the U.S.–Middle East GPR index, again holding it fixed at its February 2026 level through June. Since both counterfactuals use the same reference month, horizon, and identification strategy, their comparison isolates the implications of measuring geopolitical risk at different levels of geographical aggregation.

Figure 13 points to a substantially larger event-specific increase in geopolitical risk. Relative to its "No-War" path, U.S.–Middle East GPR rises by almost 140% in March,

Figure 13: Iran conflict: "No-War" counterfactual using U.S.–Middle East GPR




*Notes:* The left-hand panels report the observed paths of the U.S.–Middle East GPR index, real GDP, consumer prices, and the two-year Treasury yield (solid lines), together with their "No-War" counterfactual paths (orange dashed lines). The counterfactual holds the U.S.–Middle East GPR index fixed at its February 2026 level through June 2026 by allowing the identified structural GPR shock to adjust endogenously. Orange shaded areas denote 68% credible intervals. The right-hand panels report the difference between the actual and counterfactual paths. Green lines denote posterior medians and shaded areas the corresponding 68% credible intervals. Real GDP and consumer-price differences are expressed in percent, and Treasury-yield differences in percentage points.

roughly twice the corresponding increase in the aggregate index, and remains markedly elevated through June. The implied output effects are also somewhat stronger. The GDP gap widens steadily after the escalation and reaches approximately $-0.4\%$ by June, compared with around $-0.3\%$ under the aggregate-GPR counterfactual. Consumer prices follow a broadly similar path in the two exercises, rising by around 0.1% relative to the counterfactual during the spring before the gap narrows. The response of the two-year Treasury yield differs more visibly: under the Middle East specification, the yield initially rises by around 0.04–0.05 percentage points relative to its "No-War" path, although the difference subsequently fades.

The comparison therefore shows that geographical disaggregation matters for the

macroeconomic effects attributed to this episode. The Middle East index captures a substantially larger increase in geopolitical risk and is associated with a larger decline in U.S. activity than the aggregate measure. This result is consistent with the impulse-response evidence in the previous section, where directed Middle East GPR shocks lead to a larger and more delayed contraction in U.S. output than aggregate GPR shocks.

# 5 Concluding remarks

This paper proposes a new approach to measuring geopolitical risk (GPR) by decomposing it into its geographical sources. Building on the benchmark methodology of Caldara and Iacoviello (2022), we construct *directed* GPR indices using local-language news sources for four major economic regions. Our methodology allows us to trace the origin of outward-looking geopolitical risks and to assess how they relate to macroeconomic and financial conditions.

We show that directed GPR indices differ significantly from the Caldara and Iacoviello (2022) index and from each other, reflecting the diversity of national narratives and threat perceptions. A substantial share of GPR coverage cannot be attributed to a particular external region, while the composition of the remaining external component varies markedly across countries and over time. Narratives involving multiple regions are also quantitatively important, underscoring the value of a mutually exclusive decomposition that avoids double-counting while preserving geopolitical interconnections. Our empirical applications show that the macroeconomic effects of GPR shocks differ in timing, magnitude, persistence, and, in some cases, sign depending on their geographical origin. The counterfactual analysis of the 2026 conflict involving Iran further demonstrates that a directed index can identify a sharper increase in geographically concentrated risk, and correspondingly, larger economic effects than an aggregate measure.

Our results connect to the growing policy and academic literature that treats geopolitical risk and geoeconomic fragmentation as relevant drivers of trade, supply-chain resilience, and financial stability. In this context, our directed, local-language GPR indices complement existing measures and provide a framework for international macro-financial analysis and policy monitoring. We make these indices available through a regularly updated companion webpage, together with a broader database of local GPR measures for 34 economies. Future work could extend the directed framework to additional countries, explore higher-frequency data, refine the classification of narratives involving multiple actors, and investigate the mechanisms underlying the heterogeneous macroeconomic responses documented in this paper.

**Declaration of generative AI and AI-assisted technologies in the writing process** During the preparation of this work the author(s) used ChatGPT for revising the draft, checking for typos and improving the readability of the text and the charts. The author reviewed and edited the output as needed and take full responsibility for the content of the published article.

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# Appendix

## A Constructing Local GPR

### A.1 Search query used in each language

Below we give the search query as used in the Factiva search engine that classifies each article as either containing such terms or not. The asterisk stands for a wildcard, that is, the presence of however many non-empty symbols followed by the end of the word. The question mark corresponds to a non-empty symbol that is between the given symbols.

**GPR Terms: English** (((war or wars or conflict* or hostilit* or revolution* or insurrection* or uprising* or revolt* or coup* or geopolitic* or terror* or guerilla* or hostage*) Near2 (threat* or warn* or fear* or risk* or concern* or danger* or doubt* or cris* or trouble* or dispute* or tension* or imminent or inevitable or menace* or brink* or scare* or peril*)) or ((peace or truce or armistice or treat* or parley) Near2 (threat* or menace or reject* or peril or boycott or disrupt*)) or ((military or troop* or missile* or arms or weapon* or bomb* or warhead*) and (buildup or build-up or sanction* or blockade* or embargo* or quarantine or ultimatum or mobiliz*)) or ((((nuclear or atomic) near1 (war* or bomb* or missile*)) or h-bomb* or thermo-nuclear or hydrogen bomb* or nuclear test* or nuclear weapon*) and (threat* or warn* or fear* or risk* or concern* or danger* or doubt* or cris* or trouble* or dispute* or tension* or imminent or inevitable or menace* or brink* or scare* or peril*)) or ((war or wars or conflict* or hostilit* or revolution* or insurrection* or uprising* or revolt* or coup* or geopolitic*) Near2 (begin* or start* or declare* or begun or began or outbreak or broke out or breakout or proclam* or launch*)) or ((allie* or enem* or insurgen* or foe* or army or navy or aerial or troop* or rebel*) Near2 (advance* or attack* or strike* or drive* or shell* or offensive* or invasion* or invad* or clash* or raid* or launch*)) or ((terror* or guerilla* or hostage*) Near2 (attack* or act* or bomb* or kill* or strike* or hijack*)))

**GPR Terms: Simplified Chinese** (((战争 or 冲突 or 敌对 or 革命 or 起义 or 造反 or 暴动 or 叛乱 or 反叛 or 政变 or 地缘政治 or 恐怖 or 游击 or 人质) Near2 (威胁 or 警告 or 恐惧 or 风险 or 忧虑 or 岌岌可危 or 担忧 or 危险 or 疑虑 or 危机 or 困境 or 争端 or 紧张 or 迫在眉睫 or 不可避免 or 边缘 or 恐慌 )) or ((和平 or 休战 or 停战 or 停战协议 or 条约 or 谈判 or 会谈) Near2 (威胁 or 否决 or 拒绝 or 岌岌可危 or 抵制 or 扰乱 or 破坏)) or ((军事 or 部队 or 导弹 or 武器 or 炸弹 or 弹头) and (集结 or 增强 or 升级 or 制裁 or 封锁 or 禁运 or 隔离 or 最后通牒 or 动员)) or ((((核 or 原子) near1 (战争 or 弹 or 导弹)) or 氢弹 or 热核 or 核试验 or 核武器) and (威胁 or 警告 or 恐惧 or 风险 or 忧虑 or 担忧 or 危险 or 疑虑 or 危机 or 困境 or 争端 or 紧张 or 迫在眉睫 or 不可避免 or 边缘 or 恐慌 or 岌岌可危)) or ((战争 or 冲突 or 敌对 or 革命 or 起义 or 造反 or 暴动 or 叛乱 or 反叛 or 政变 or 地缘政治 or 恐怖 or 游击 or 人质) Near2 (开始 or 发动 or 宣布 or 已经开始 or 开始了 or 爆发 or 宣告)) or (((盟友 or 敌军 or 叛乱分子 or 敌人 or 陆军 or 军队 or 海军 or 空中 or 部队 or 叛军)

Near2 (推进 or 攻击 or 袭击 or 打击 or 前进 or 炮击 or 进攻 or 入侵 or 冲突 or 突袭 or 发射) or 空袭)) or ((恐怖 or 游击 or 人质) Near2 (袭击 or 行为 or 爆炸 or 杀害 or 打击 or 劫持)))

**GPR Terms: Russian** (((войн* or воен* or боев* or бой or бои or конфликт* or вражд* or революци* or восстани* or бунт* or переворот* or геополити* or террор* or партизан* or заложник* or столкновени* or противостояни* or конфронтаци*) Near2 (угроз* or предупрежден* or предостер* or боят* or бояз* or риск* or опасен* or опасн* or сомнен* or кризис* or проблем* or спор* or напряжен* or неминуем* or неизбежн* or трево*)) or ((мир or перемир* or договор* or переговор*) Near2 (угроз* or отклон* or опасност* or бойкот* or нарушен*)) or ((военн* or войск* or ракет* or оружие* or бомб* or боеголовк*) and (накоплен* or санкци* or блокад* or эмбарго* or карантин or ультиматум or мобилиз*)) or ((((ядерн* or атомн*) near1 (войн* or бомб* or ракет*)) or водородн* бомб* or термоядерн* or водородн* бомб* or ядерн* испытан* or ядерн* оружи*) and (угроз* or предупрежден* or предостер* or боят* or бояз* or риск* or опасен* or опасн* or сомнен* or кризис* or проблем* or спор* or напряженност* or неминуем* or неизбежн*)) or ((войн* or воен* or боев* or бой or бои or конфликт* or вражд* or революци* or восстани* or бунт* or переворот* or геополити* or столкновен*) Near2 (начин* or начал* or объяв* or начат* or вспышк* or вспых* or провозглаш* or эскал* or разжига* or обостр*)) or ((союзник* or враг* or повстанц* or противник* or арми* or флот* or войск* or повстанец* or мятежник* or сепаратист*) Near2 (продви* or атак* or удар* or наступлен* or обстрел* or наступат* or вторжен* or вторга* or столкновен* or рейд* or запущен*)) or ((террор* or партизан* or заложник* or боевик* or мятежник* or сепаратист*) Near2 (атак* or действ* or бомб* or убийств* or удар* or угон* or нападен* or захва* or акци* or акт*)) or теракт*)

**GPR Terms: German** (((Krieg* or Konflikt* or Feindseligkeit* or Revolution* or Aufstand or Aufstände or Erhebung* or Revolte* or Putsch* or geopolitisch* or Terror* or Guerilla* or Geisel*) Near3 (Bedrohung* or warn* or Angst or Ängste or Risik* or Bedenken* or Gefahr* or Zweifel* or Krise* or Ärger* or Streit* or Spannung* or unmittelbar* or unvermeidlich* or Bedrohung* or Rand or Schrecken* or Gefahr*)) or ((frieden or waffenstillstand or waffenstillstände or waffenruhe or vertrag or verträge or gespräch*) Near3 (bedrohung* or ablehnen* or gefahr* or boykott* or stören*)) or ((militär or truppe* or rakete* or waffen or waffe* or bombe* or sprengkopf*) and (aufbau or sanktion* or blockade* or embargo* or quarantäne or ultimatum or mobilisierung*)) or ((((nuklear or atomar) near1 (krieg* or bombe* or rakete*)) or H-Bombe* or thermonuklear or Wasserstoffbombe* or Nukleartest* or Nuklearwaffe*) and (Bedrohung* or warn* or Angst* or Risiko* or Bedenken* or Gefahr* or Zweifel* or Krise* or Ärger* or Streit* or Spannung* or unmittelbar* or unvermeidlich* or Bedrohung* or Rand* or Schrecken* or Gefahr*)) or ((Krieg* or Konflikt* or Feindseligkeit* or Revolution* or Aufstand or Aufstände or Erhebung* or Revolte* or Putsch* or geopolitisch*) Near3 (beginnen* or starten* or erklären* or begann or begonnen or Ausbruch or ausgebrochen or Proklamation* or start*)) or ((Verbündet* or Feind* or Aufständisch* or Gegner* or Armee* or Marine or Luft* or Trupp* or Rebell*) Near3 (vorstoßen* or vorstöße or angreifen* or schlagen* or

treiben* or beschießen* or Offensive* or Invasion* or einmarschieren* or Zusammenstöße or Zusammenstoß or Überfall or Überfälle or starten*)) or ((Terror* or Guerilla* or Geisel*) Near3 (angriff* or akt* or bombe* or töten* or schlag* or entführung*)))

**GPR Terms: French** (((guerre* or conflit* or hostilité* or révolution* or insurrection* or soulèvement* or révolte* or coup d'état* or géopolitique or terror* or guérilla* or otage*) Near3 (menace* or avert* or peur* or risque* or préoccupation* or danger* or doute* or crise* or problème* or dispute* or tension* or imminent or inévitable or menace* or bord* or effrayer* or péril*)) or ((paix or trêve or armistice or trait* or pourparl*) Near3 (menace* or rejet* or péril* or boycott* or perturb*)) or ((militaire* or troupe* or missile* or armes or arme* or bombe* or ogive*) and (accumulation or sanction* or blocus* or embargo* or quarantaine or ultimatum or mobilisa*)) or ((((nucléaire* or atomique*) near1 (guerre* or bombe* or missile*)) or bombe* H or thermo-nucléaire* or bombe* à hydrogène or essai* nucléaire* or arme* nucléaire*) and (menace* or avert* or peur* or risque* or préoccupation* or danger* or doute* or crise* or problème* or dispute* or tension* or imminent or inévitable or menace* or bord* or effrayer* or péril*)) or ((guerre* or conflit* or hostilité* or révolution* or insurrection* or soulèvement* or révolte* or coup d'état* or géopolitique) Near3 (commenc* or début* or déclar* or éclat* or proclama* or lancer*)) or ((allié* or ennemi* or insurgé* or adversaire or armée or marine* or aérien* or troupe* or rebelle*) Near3 (avanc* or attaq* or frapp* or pouss* or bombard* or offensive* or invasion* or envahir* or affrontement* or raid* or lanc*)) or ((terror* or guérilla* or otage*) Near3 (attaqu* or acte* or bombe* or tue* or frappe* or détournement*)))

**GPR Terms: Italian** (((guerr* or conflitt* or ostilità or rivoluzion* or insurrezion* or sollevament* or rivolt* or colp* di stato or geopolitic* or terror* or guerrigli* or ostaggi*) Near3 (minacc* or avvert* or paur* or rischi* or preoccupazion* or pericol* or dubbi* or crisi or problem* or disput* or tension* or imminent* or inevitabil* or orlo or spavent* or pericol*)) or ((pace or tregua or armistizio or trattat* or parlament*) Near3 (minacc* or rifiut* or pericolo or boicott* or disturb*)) or ((militare or trupp* or missil* or armi or arma or bomb* or testata) and (accumulo or sanzion* or blocc* or embargo or quarantena or ultimatum or mobilitazion*)) or ((guerr* or conflitt* or ostilità or rivoluzion* or insurrezion* or sollevament* or rivolt* or colpo di stato or geopolitic*) Near3 (inizia* or comincia* or dichiara* scoppio or scoppia* or proclama* or lancia* or dichiarazione)) or ((alleat* or nemic* or insort* or avversari* or esercit* or marina or aere* or truppa or ribell*) Near3 (avanza* or attacca* or colpi* or spinge* or bombarda* or offensiv* or invasion* or invade* or scontr* or raid or lancia*)) or ((terror* or guerrigli* or ostaggi*) Near3 (attacc* or atto or atti or bomb* or uccide* or colpo* or dirottament*)))

**GPR Terms: Spanish** ((((guerra* or conflicto* or hostilidad* or revoluci?n* or insurrecci?n* or levantamiento* or revuelta* or golpe* de estado or geopolít?c* or terror* or guerrilla* or reh?n*) Near3 (amenaza* or advert* or miedo* or riesgo* or

preocupaci?n* or peligro* or duda* or crisis or problema* or disputa* or tensi?n* or inminente* or inevitable* or borde* or susto* or peligro*)) or ((paz or tregua or armisticio or trat*) Near3 (amenaz* or rechaz* or peligro* or boicot* or perturb*)) or ((militar* or tropa* or misil* or armas or arma* or bomba* or ojiva*) and (acumulaci?n* or sanci?n* or bloqueo* or embargo* or cuarentena or ultimátum or moviliza*)) or ((((nuclear* or atómic*) near1 (guerra* or bomba* or misil*)) or bomba* H or termonuclear* or bomba* de hidrógeno or prueba* nuclear* or arma* nuclear*) and (amenaza* or advert* or miedo* or riesgo* or preocupaci?n* or peligro* or duda* or crisis or problema* or disputa* or tensi?n* or inminente* or inevitable* or amenaza* or borde* or susto* or peligro*)) or ((guerra* or conflicto* or hostilidad* or revoluci?n* or insurrecci?n* or levantamiento* or revuelta* or golpe* de estado or geopolític*) Near3 (comenz* or empez* or declar* or brote* or estallido or proclam* or lanz*)) or ((aliad* or enemig* or insurgent* or adversario* or ejército or marina or aéreo* or tropa* or rebelde*) Near3 (avanz* or atac* or golp* or impuls* or bombard* or ofensiva* or invasi?n* or invad* or choque* or redada* or lanz*)) or ((terror* or guerrilla* or reh?n*) Near3 (ataque* or acto* or bomba* or mat* or golpe* or secuestr*))) not (sin embargo))

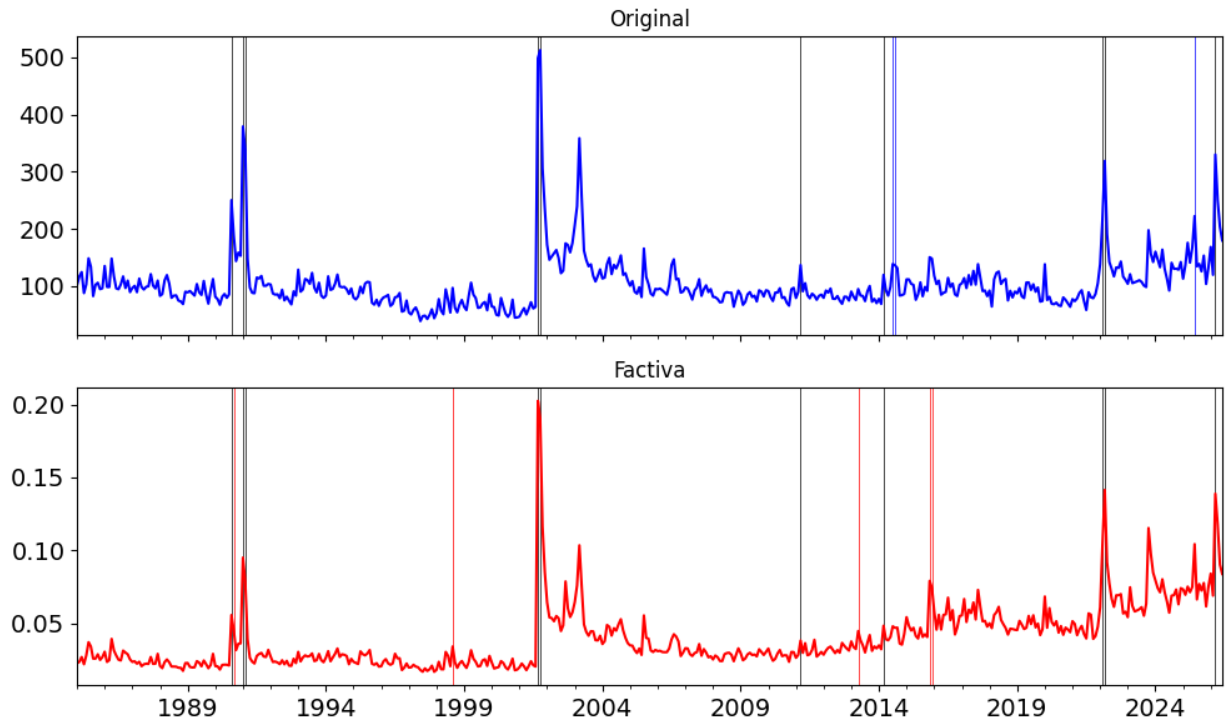
# B Reconstructing C&I GPR Index with Factiva

In this section we show that Factiva is able to sufficiently accurately reproduce the original C&I GPR index. The aim of this exercise is to understand which differences between our indices come from the specific news repository, and which come from the more fundamental methodological difference of using local sources.

Figure B.1 compares the original Caldara and Iacoviello (2022) (C&I) GPR index[20], based on the Anglo-Saxon press, with our Factiva-based reproduction. The two indices exhibit very similar dynamics, with a correlation coefficient of 0.82. Table B.1 complements this comparison by reporting the incidence of major geopolitical events across the two series. Most of these events are associated with increases of at least two standard deviations in both indices.

As an additional validation exercise, Figure B.2 reports the effects of a GPR shock on the U.S. economy using the same specification as in Caldara and Iacoviello (2022). The impulse responses obtained with our Factiva-based index are virtually identical to those obtained with the original C&I index.

Figure B.1: GPR (English-language press): original index (C&I) and replication with Factiva



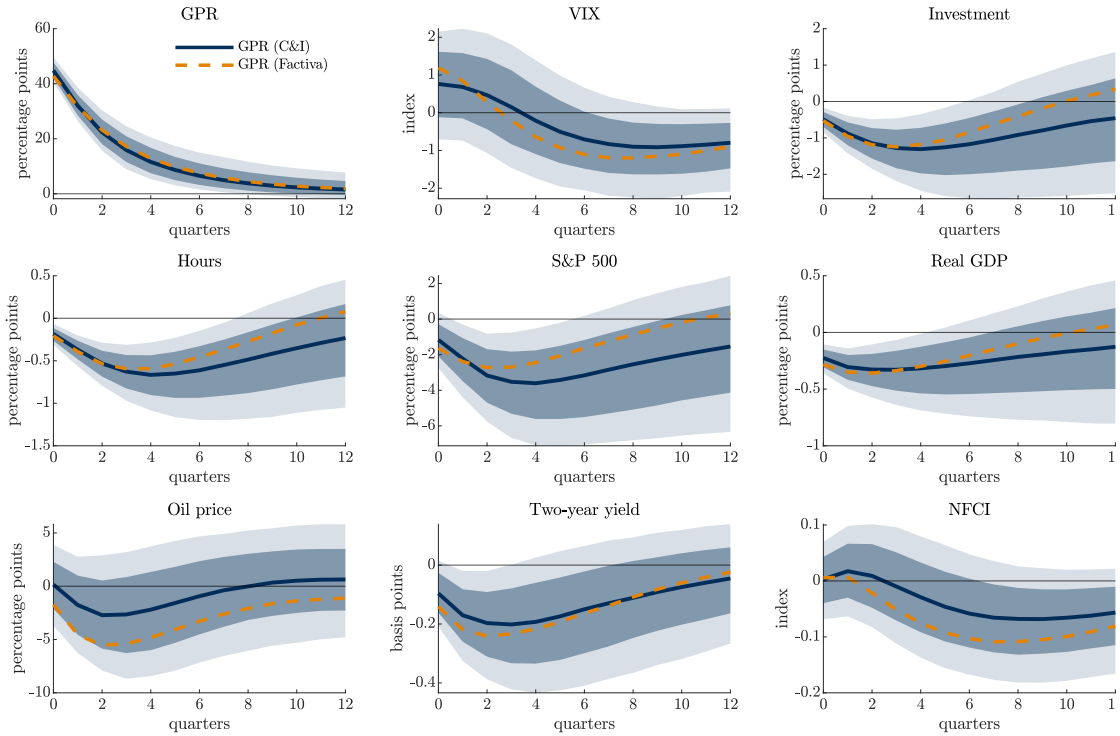

*Notes*: Vertical lines mark the peaks, defined as values over three standard deviations away from the mean, using a rolling window of 36 months (following C&I). Both series are relative article counts normalized to 100 in the period 1985-2019. The top subplot shows the original GPR index as constructed by Caldara and Iacoviello (2022), and obtained from https://www.matteoiacoviello.com/gpr.htm, while the bottom plot displays our reproduction using Factiva.

[20] Available for download at https://www.matteoiacoviello.com/gpr.htm.

Table B.1: GPR (English-language press): Narrative of events in original index and Factiva replication

| | | Index Type | | | |
|---|---|---|---|---|---|
| **Date** | **Event Description** | 3 SD | 2 SD | 1 SD | 0.5 SD |
| **Aug 1990** | Iraq invades Kuwait | Both | Both | Both | Both |
| **Sep 1990** | Continued tensions (Iraq) | Factiva | Both | Both | Both |
| **Jan-Feb 1991** | The First Gulf War | Both | Both | Both | Both |
| **Aug 1998** | The US embassies bombings in Kenya and Tanzania by Al-Qaeda | Factiva | Both | Both | Both |
| **Sep-Oct 2001** | 9/11 | Both | Both | Both | Both |
| **Mar 2011** | Military interventions in Libya | Both | Both | Both | Both |
| **Apr 2013** | The Boston Marathon bombing | Factiva | Factiva | Factiva | Both |
| **Mar 2014** | Russia annexes Crimea | Both | Both | Both | Both |
| **Jul-Aug 2014** | The downing of MH17, Israel-Gaza conflict, fighting in Ukraine | Original | Both | Both | Both |
| **Nov 2015** | The Paris terrorist attacks | Factiva | Both | Both | Both |
| **Dec 2015** | The San Bernardino terrorist attack | Factiva | Both | Both | Both |
| **Jan 2022** | Rumours about upcoming invasion | Original | Original | Both | Both |
| **Feb -Mar 2022** | Russia invades Ukraine | Both | Both | Both | Both |
| **June 2025** | Israeli-Iran conflict | Original | Both | Both | Both |
| **March 2026** | US-Israel-Iran war | Both | Both | Both | Both |

Figure B.2: The macroeconomic effects of GPR shocks on the U.S. economy

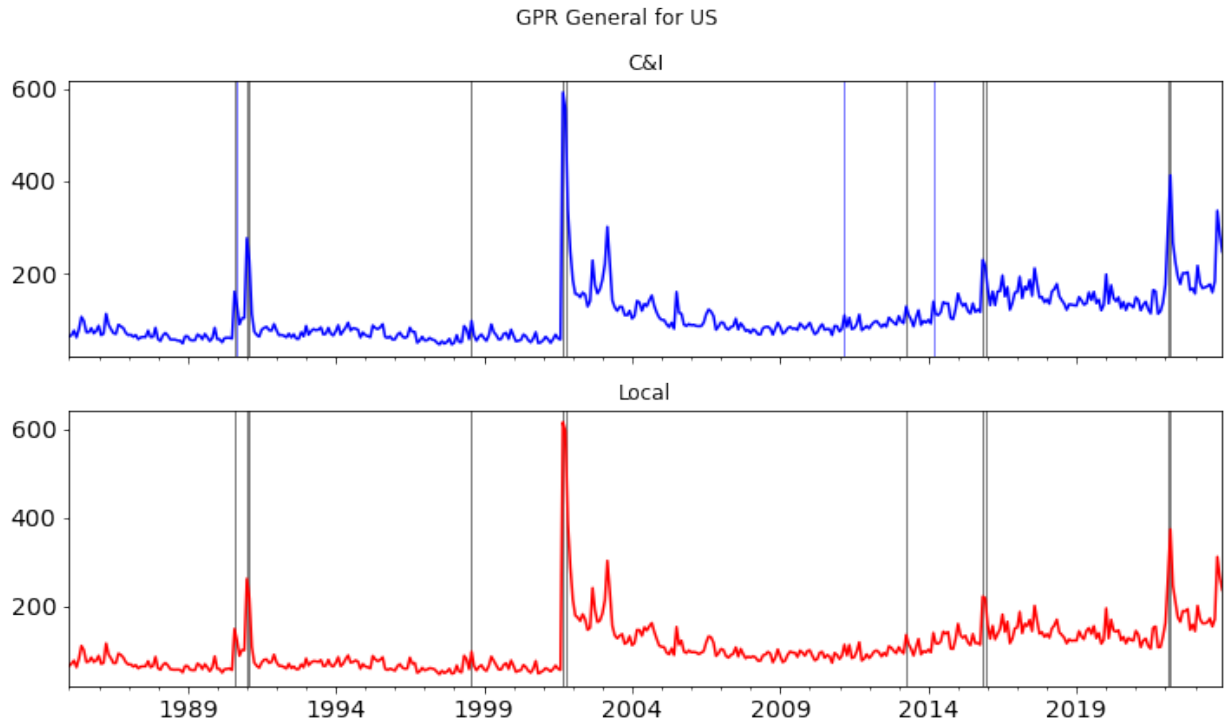



*Notes:* The solid line shows the median impulse response of the indicated variable to a two-standard-deviation increase in the GPR index of Caldara and Iacoviello (2022, C&I). The dark and light shaded areas denote the 68% and 90% credible sets, respectively. The orange dashed line shows the corresponding median response to an equivalent increase in our Factiva-based GPR index. The sample period is 1986Q1–2019Q4.

# C Local GPR narratives, and their comparison with C&I GPR Index narrative

The purpose of the following Figures C.3–C.5 and Tables C.2–C.4 is two-fold. First, they function as narrative validation of the GPR indices of countries: U.S., Russia, and China. Second, they can be used as comparison for the types of events that make it to the local indices and not to the general C&I one (and vice versa). The respective narratives behind the European indices used in this paper can be found in Alonso-Alvarez et al. (2026).

Figure C.3: US GPR

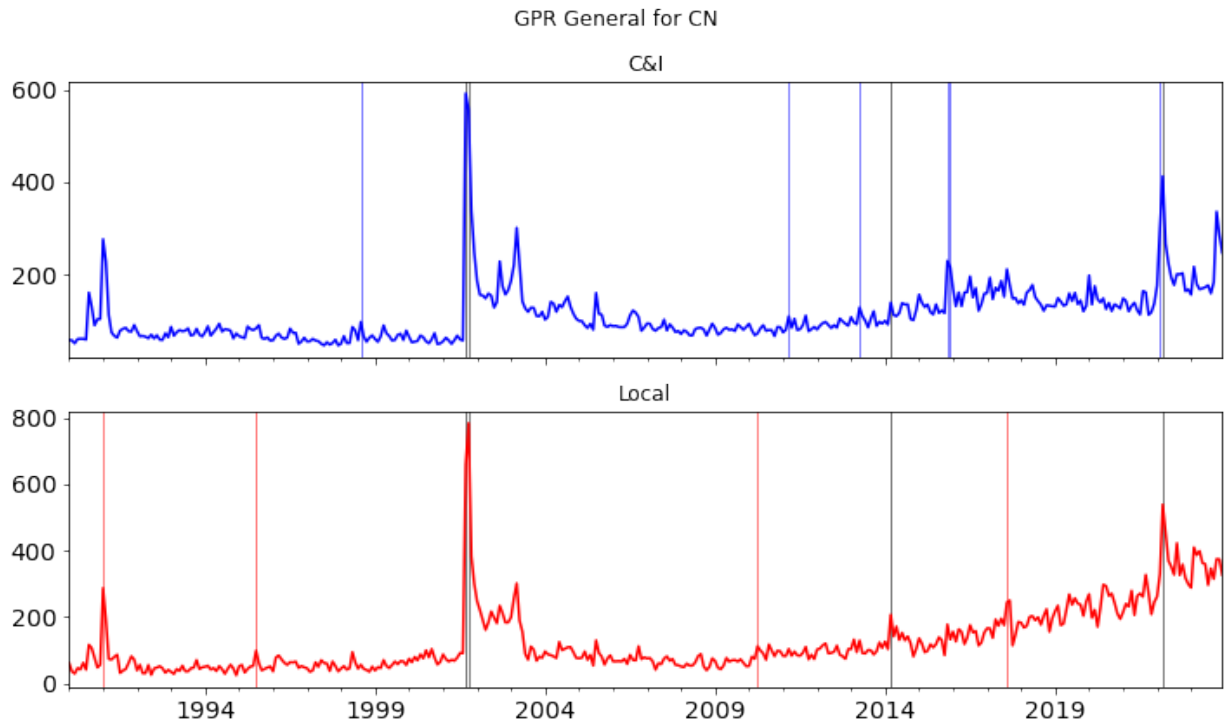



Notes: Vertical lines mark the peaks defined as values over 3 standard deviations away from the mean, using a rolling window of 36 months. Both series are relative article counts normalized to 100 in the period 1985-2019. The top subplot shows the GPR index constructed using the sources of C&I (the equivalent of the original GPR index but reproduced in Factiva); the bottom plot displays the GPR index constructed using the local sources of the country.

Table C.2: Narrative of events of C&I GPR (English-speaking press) versus US media (local)

| | | Index Type | |
|---|---|---|---|
| **Date** | **Event Description** | 3 SD | 2 SD |
| **Aug 1990** | Iraq invades Kuwait (C) | Both | Both |
| **Sep 1990** | Continued tensions (Iraq) (C) | C&I | Both |
| **Jan - Feb 1991** | The First Gulf War (B) | Both | Both |
| **Aug 1998** | The US embassies bombings in Kenya and Tanzania by al-Qaeda (B) | Both | Both |
| **Sep - Oct 2001** | 9/11 (D) | Both | Both |
| **Mar 2011** | Military intervention in Libya (B) | C&I | Both |
| **Apr 2013** | The Boston Marathon bombing (D) | Both | Both |
| **Mar 2014** | Russia annexes Crimea (C) | C&I | Both |
| **Nov 2015** | The Paris terrorist attacks (F) | Both | Both |
| **Dec 2015** | The San Bernardino terrorist attack (US) (D) | Both | Both |
| **Feb - Mar 2022** | Russia invades Ukraine (A) | Both | Both |

Figure C.4: China GPR

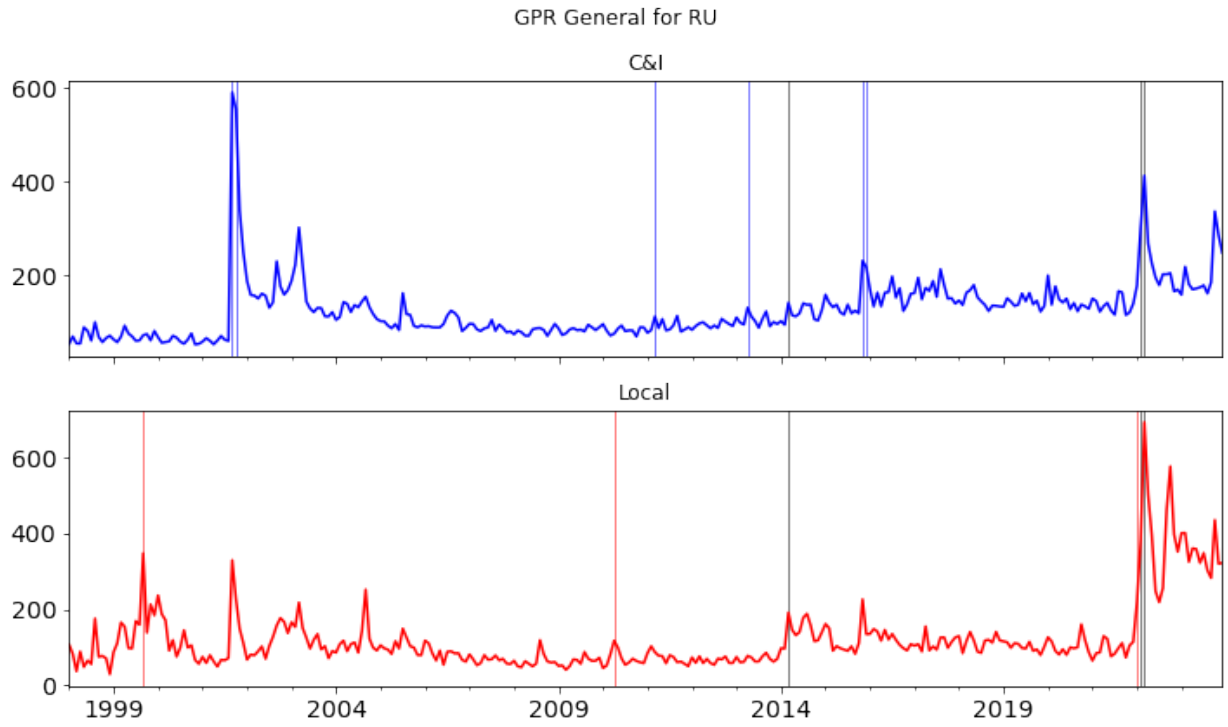



Notes: Vertical lines mark the peaks defined as values over 3 standard deviations away from the mean, using a rolling window of 36 months. Both series are relative article counts normalized to 100 in the period 1985-2019. The top subplot shows the GPR index constructed using the sources of C&I (the equivalent of the original GPR index but reproduced in Factiva); the bottom plot displays the GPR index constructed using the local sources of the country.

Table C.3: Narrative of events of C&I GPR (English-speaking press) versus Chinese media (local)

| | | Index Type | | | |
|---|---|---|---|---|---|
| **Date** | **Event Description** | 3 SD | 2 SD | 1 SD | 0.5 SD |
| **Jan 1991** | The First Gulf War (C) | Local | Both | Both | Both |
| **Jul 1995** | Third Taiwan Strait Crisis (announcement) (B) | Local | Local | Local | Both |
| **Aug 1998** | The US embassies bombings in Kenya and Tanzania by al-Qaeda (F) | C&I | C&I | C&I | C&I |
| **Sep - Oct 2001** | 9/11 (A) | Both | Both | Both | Both |
| **Apr 2010** | - | Local | Local | Local | Local |
| **Mar 2011** | Military intervention in Libya (C) | C&I | C&I | Both | Both |
| **Apr 2013** | The Boston Marathon Bombing (F) | C&I | Both | Both | Both |
| **Mar 2014** | Russia annexes Crimea (C) | Both | Both | Both | Both |
| **Nov 2015** | The Paris terrorist attacks (F) | C&I | Both | Both | Both |
| **Dec 2015** | The San Bernadino terrorrist attack (US) (F) | C&I | C&I | C&I | C&I |
| **Aug 2017** | The Doklam standoff between China and India in the Himalayas (B) | Local | Local | Both | Both |
| **Feb 2022** | Russia invades Ukraine (A) | C&I | Both | Both | Both |
| **Mar 2022** | Recent invasion of Ukraine by Russia (A) | Both | Both | Both | Both |

Figure C.5: Russia GPR




Notes: Vertical lines mark the peaks defined as values over 3 standard deviations away from the mean, using a rolling window of 36 months. Both series are relative article counts normalized to 100 in the period 1985-2019. The top subplot shows the GPR index constructed using the sources of C&I (the equivalent of the original GPR index but reproduced in Factiva); the bottom plot displays the GPR index constructed using the local sources of the country.

Table C.4: Narrative of events of C&I GPR (English-speaking press) versus Russian media (local)

| | | Index Type | | | |
|---|---|---|---|---|---|
| **Date** | **Event Description** | 3 SD | 2 SD | 1 SD | 0.5 SD |
| **Sep 1999** | Moscow terrorist bombings (D) | Local | Local | Local | Local |
| **Sep 2001** | 9/11 (A) | C&I | Both | Both | Both |
| **Oct 2001** | 9/11 (A) | C&I | C&I | Both | Both |
| **Apr 2010** | Moscow metro terrorist attack (D) | Local | Local | Local | Local |
| **Mar 2011** | Military intervention in Libya (C) | C&I | C&I | C&I | Both |
| **Apr 2013** | The Boston Marathon bombing (F) | C&I | C&I | C&I | Both |
| **Mar 2014** | Russia annexes Crimea (B) | Both | Both | Both | Both |
| **Nov 2015** | The Paris terrorist attacks (F) | C&I | Both | Both | Both |
| **Dec 2015** | The San Bernardino terrorist attack (US) (F) | C&I | C&I | C&I | Both |
| **Jan 2022** | Rumours about upcoming invasion (D) | Local | Local | Both | Both |
| **Feb - Mar 2022** | Russia invades Ukraine (B) | Both | Both | Both | Both |

# D Factiva's filter and zoning robustness

We now present the background statistical analysis carried out to ensure robustness of our directed GPR indices, specifically with regards to the "re" classification of Factiva, which it uses to assign the focal geography of the article (there are usually several of such tags associated with each article, as well as their nested equivalents - e.g. having both "France" and "Europe"). For directed indices, the main issue is whether Factiva is consistent in applying this classification across time - what we want to avoid is to have large parts of the database unclassified.

To illustrate our logic, we will go through the exercise for Russia, and report the results of the same exercise for the three other countries in Table D.1. Results for European countries can be found in Alonso-Alvarez et al. (2026), where directed indices are called "bilateral".

First, we make a guess about the time period during which we are likely to have good Factiva filter coverage. Our first guess will correspond to the time period for which we have good newspaper coverage, which for Russia starts in 1999. We restrict all Factiva queries to this range (finishing at the end of 2023), so the results of this robustness exercise will be valid for this period. We then submit to Factiva the query to retrieve the number of articles corresponding to the GPR numerator. For Russia, there are 64248 articles that contribute to the GPR index. We then add to the query the condition that it feature at least one "re" classification, with our possible list of regions spanning the entire globe. It can be seen that for Russia almost no articles are left uncategorized. For the filter to be considered as valid, we require at least 95% of the articles to be categorized.

Note that this includes the US from 2003, and than including earlier articles from US newspapers would drastically drive down the number. This reflects an important idiosyncrasy in Factiva´s coverage of the US: early US newspapers are inconsistently categorised in terms of region of focus. Our solution to this is as follows: when the use of the filter is vital (as it will be in the Decomposition exercise), we will restrict our U.S. results to start from 2003. However, when considering the straightforwards marginal directed GPR, for example US->Russia, we adopt a different approach:

To obtain directed GPRs for US->Russia, US->Middle East, and US->China, we extend the US query with a list of countries/capitals/relevant terms that would stand for the relevant region. We try several variations, and pick one that has the highest correlation with the standard "re"-based directed GPR (at least 0.98) *on the timespan where that GPR is valid, that is, 2003 onward*. We then use these "term-based" directed GPRs when considering directed US GPRs. These indices will span the whole length of the US GPR time-series, that is, back to 1985. Specifically, we use the following additions:

- US->RU: (atleast4 Russia* or atleast4 Moscow* or atleast4 USSR* or atleast4 Soviet* or atleast4 Gorbachev* or atleast4 Yeltsin* or atleast4 Putin* or atleast4 Medvedev*)
- US->ME: (atleast4 Israel* or atleast4 Palestin* or atleast4 Iraq* or atleast4 Iran* or atleast4 Afghanistan* or atleast4 Leban* or atleast4 Turk* or atleast4 Syria* or atleast4 Egypt* or atleast4 Saudi Arab* or atleast4 Yemen* or atleast4 Kurd* or atleast4 Jordan* or atleast4 Gaza* or atleast4 Tunis* or atleast4 Libya* or (ISIL or ISIS or Daesh or Islamic State))
- US->CN: (atleast4 Chinese* or atleast4 China*)

Table D.1: Article counts behind the “re" Factiva categorization

| | **US (1985-2002)** | **US (2003-2023)** | **RU (1999-2023)** | **CN$_{en}$ (1995-2023)** | **CN$_{zhcn}$ (1995-2023)** |
|---|---|---|---|---|---|
| GPR | 335188 | 564308 | 64248 | 80694 | 205575 |
| Categorised | 201803 | 551910 | 63963 | 79039 | 195936 |
| Fraction categorised | 0.60 | 0.98 | 1.00 | 0.98 | 0.95 |

Note: Queries are defined by modifying the local GPR queries with extra regional sets. Local GPR queries are given in A, where the query for DE does NOT have the component related to Nuclear war (we also add to each query the list of newspapers associated with each country of origin). The “Categorised" query contains, in addition to the above, the following specification: *and (re=(ASIAZ or apacz or ausnz or USA or CANA or EURZ or africaz or caribz or lamz) or ISIS or Islamic State)*, where “ISIS" and “Islamic state" is given in the respective language of the country of origin. This is meant to represent all the possible geographies covered by Factiva, using the most broad filter categories.

# E *Internal* and *external* GPR components

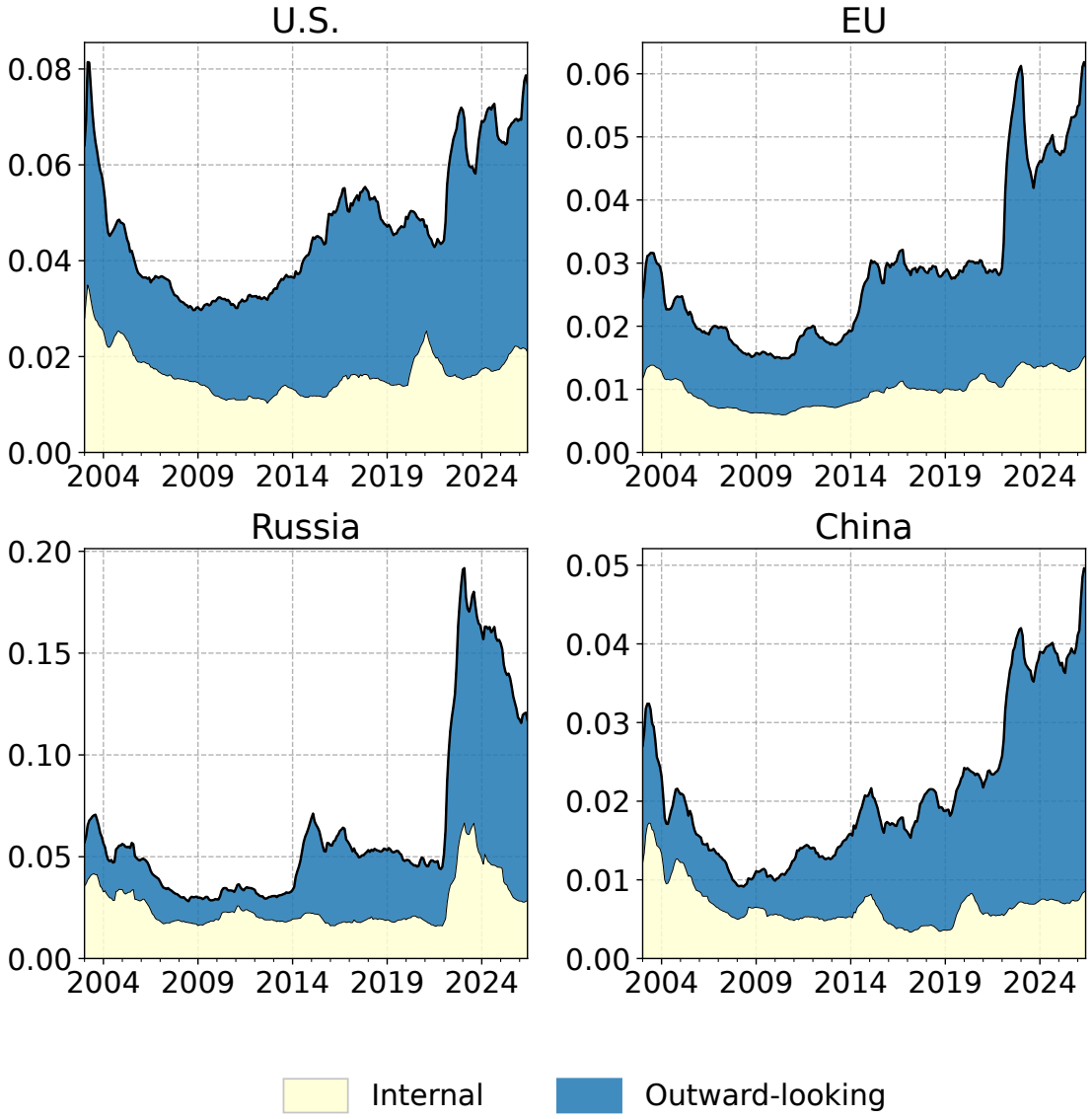
We first detail our method for using the focal region “re" filter to identify internal and external GPR component for any given country, as defined by a query, a language and a set of sources in that language. First, we need to ensure validy of the filter for the given time period - the process describing it is detailed in Appendix D. This process gives us the articles that are tagged with “re" (the final line of Table D.1). We then compose a logical query that covers only those articles that contain a reference to *any other region than the country of origin*. We call this the “Other countries" query. For Russia, for instance, there are 37970 such articles. The key assumption we make is that these are the articles that make up the External part of the GPR; that the other articles will only have substantial references to Russia, and will make no, or very little, mentions of other countries or regions. Since we know that the internal noise forms a significant and non-negligible part of the GPR index, we are not concerned that this number is smaller than the total GPR number. What we do want to ensure is that the query corresponding to such “other" regions is as extensive as possible, so as not to undercount the contribution to the external part of the GPR, and by extension misclassify truly external articles as internal.

Figure E.1 shows graphically the prevalence and change in the internal component of the GPR of the four economic regions considered.

We now list some of the reasons GPR-relevant articles might not be associated to region external to the country of origin. This list is a summary of manually going through a random selection of such internal articles.

- Use of geopolitical risk terms in unrelated contexts. Examples would be articles mentioning mobilizing the state police because a snow storm bombards the area; or articles talking about quarantines and looking for weapons against anxiety. The existence of these types of articles is an natural limitation of the dictionary methodology, and is impossible to avoid.
- Genuine internal acts of violence such as terrorism attacks. This part is the more genuine geopolitical risk that is truly internal to the country in question. Here articles would be talking about the acts, the perpetrators being sentenced, or any consequence suffered as a result of it. For example, a memorial service for the victims; or new measures put in place to prevent more attacks,

Figure E.1: Internal v External component in the GPR




Notes: Each panel decomposes the GPR index constructed from a country's local news sources into *external* (top) and *internal* (bottom) components. The *external* component comprises GPR articles focused on at least one foreign country or region, as identified by Factiva's regional classification. The *internal* component comprises the remaining GPR articles, which focus on domestic narratives or discuss foreign regions without sufficient emphasis to receive an external classification. Both components are expressed as shares of all news articles in the corresponding country sample and are stacked to sum to the local GPR index.

like selective passport checks.

- Passing mentions. Given the number of wars or terrorist attacks suffered under the countries in question, a large fraction of articles will simple reference them without going into detail. Examples would be: an article about the aviation business, briefly mentioning that the sector has been hurt by

terrorist attacks; an article about Brent falling, mentioning the fact that in October 2022, at the start of the Israel-Hamas conflict, Brent prices had risen (here even though a specific region is mentioned, there is not enough emphasis on it, and as such it does not get picked up by the “re” filter).

- General narrative. A number of countries have incorporated geopolitical narratives into their identity, or at least make multiple references to them in daily lives. Examples would be the War on Terror in the US, which can come up in any context, for example an article about next year´s budget. In Russia specifically the invasion functions as a time placeholder, and as such it is sometimes referenced in one way or another: for example, in an article about the local government mentioning that the last time it was changed was before the start of the "Special military operation". Additionally for Russia, another source of articles in this category would be memorial services for other wars. A large part of Russian identity is centered around the second World War, and references to it abound in the national narrative.
- General speculation: an article saying that this year retail prices fell because of “geopolitical concerns”. This encompasses speculations about the future, mentioned in passing, about possible escalations of existing conflicts, or any new terrorist attacks. Even though individual countries or regions could be mentioned, the article is not attributed to any particular one, since that would not be its main focus.
- Methodological: there is a small number of regions that are excluded from our search terms for reasons of query sizes. Sometimes smaller economies make an appearance, but we are forced to not count them as belonging to the group of "other countries", which we have to specify explicitly.

# F Decomposing *external* GPR

Table F.1: Regional Composition for the U.S., China, Russia and EU

| Region | Constituents | FACTIVA codification |
|---|---|---|
| **United States (7)** | | |
| ***Middle East*** | Cyprus, Egypt, Israel, Jordan, Kurdistan, Lebanon, Palestine, Persian Gulf countries, Syria, Turkey, Yemen, Afghanistan, Islamic State (ISIS) | re=(nafrz or meastz or AFGH) or ISIS or Islamic State |
| ***Asia Other*** | Japan, India, Pakistan, Malaysia, Indonesia, Thailand, South Korea | re=(SKOREA or INDIA or PAKIS or MALAY or INDON or THAIL or JAP) |
| ***Western Bloc*** | Canada, European Union, Nordic countries, United Kingdom | re=(CANA or EECZ or NORDZ or UK) |
| ***Russia*** | Russia | re=(RUSS) |
| ***China*** | China | re=(CHINA) |
| ***Africa*** | African countries excluding North Africa | (re=africaz not re=nafrz) |
| ***Latin America*** | Mexico, South America and the Caribbean | (re=(caribz or lamz)) |
| **China (5)** | | |
| ***Middle East*** | Cyprus, Egypt, Israel, Jordan, Kurdistan, Lebanon, Palestine, Persian Gulf countries, Syria, Turkey, Yemen, Afghanistan, Islamic State (ISIS) | re=(nafrz or meastz or AFGH) or ISIS or Islamic State |
| ***Asia Other*** | India, Pakistan, Malaysia Indonesia, Thailand | re=(INDIA or PAKIS or MALAY or INDON or THAIL) |
| ***Western Bloc*** | United States, European Union, Nordic countries, Canada, Australia, New Zealand, United Kingdom | re=(CANA or USA or EECZ or NORDZ or UK or AUSNZ) |
| ***Russia*** | Russia | re=(RUSS) |
| ***China Neighbours*** | Japan, North Korea, Taiwan, South Korea | re=(NKOREA or TAIWAN or SKOREA or JAP) |
| **Russia (5)** | | |
| ***Middle East*** | Cyprus, Egypt, Israel, Jordan, Kurdistan, Lebanon, Palestine, Persian Gulf countries, Syria, Turkey, Yemen, Afghanistan, Islamic State (ISIS) | re=(nafrz or meastz or AFGH) or Исламское or Государство or Даеш or ИГИЛ |
| ***Asia Other*** | Japan, India, Pakistan, Malaysia, Indonesia, Thailand, South Korea | re=(SKOREA or INDIA or PAKIS or MALAY or INDON or THAIL or JAP) |
| ***Western Bloc*** | Canada, United States, European Union, Nordic countries, UK | re=(CANA or USA or EECZ or NORDZ or UK) |
| ***China*** | China | re=(CHINA) |
| ***Russia Neighbours*** | Ukraine, Belorusia, Georgia, Armenia | re=(UKRN or BYELRS or GRGIA or ARMEN) |
| **France, Germany, Spain, Italy (4)** | | |
| ***Middle East*** | Cyprus, Egypt, Israel, Jordan, Kurdistan, Lebanon, Palestine, Persian Gulf countries, Syria, Turkey, Yemen, Afghanistan, Islamic State | re=(nafrz or meastz or AFGH) or Islamic State or ISIS* |
| ***China*** | China | re=CHINA |
| ***Russia*** | Russia | re=RUSS |
| ***Western Bloc*** | Canada, United States, Europe | re=(CANA or USA or FRA* or GFR* or UK or SPAIN* or ITALY* or NETH or BELG or NORDZ or IRE or SWITZ or AUST or PORL or GREECE or POL or BALTST) |

*Notes*: The number next to the country of origin refers to the number of external regions considered. For the region denominated as "Western Bloc", depending on the country of origin, that country is absent from the list of countries that make up the bloc. For China, the English-language version of the table is shown; the Simplified Chinese version has, instead of "ISIS or Islamic state", "伊斯兰国 or 达伊沙").

## F.1 Zoning robustness

We now explain the background statistical analysis carried out to ensure robustness of our decomposition, specifically with regards to the zoning. The main question here is whether the particular set of regions considered for each country provides a good enough coverage of the External, outward-looking part of the GPR.

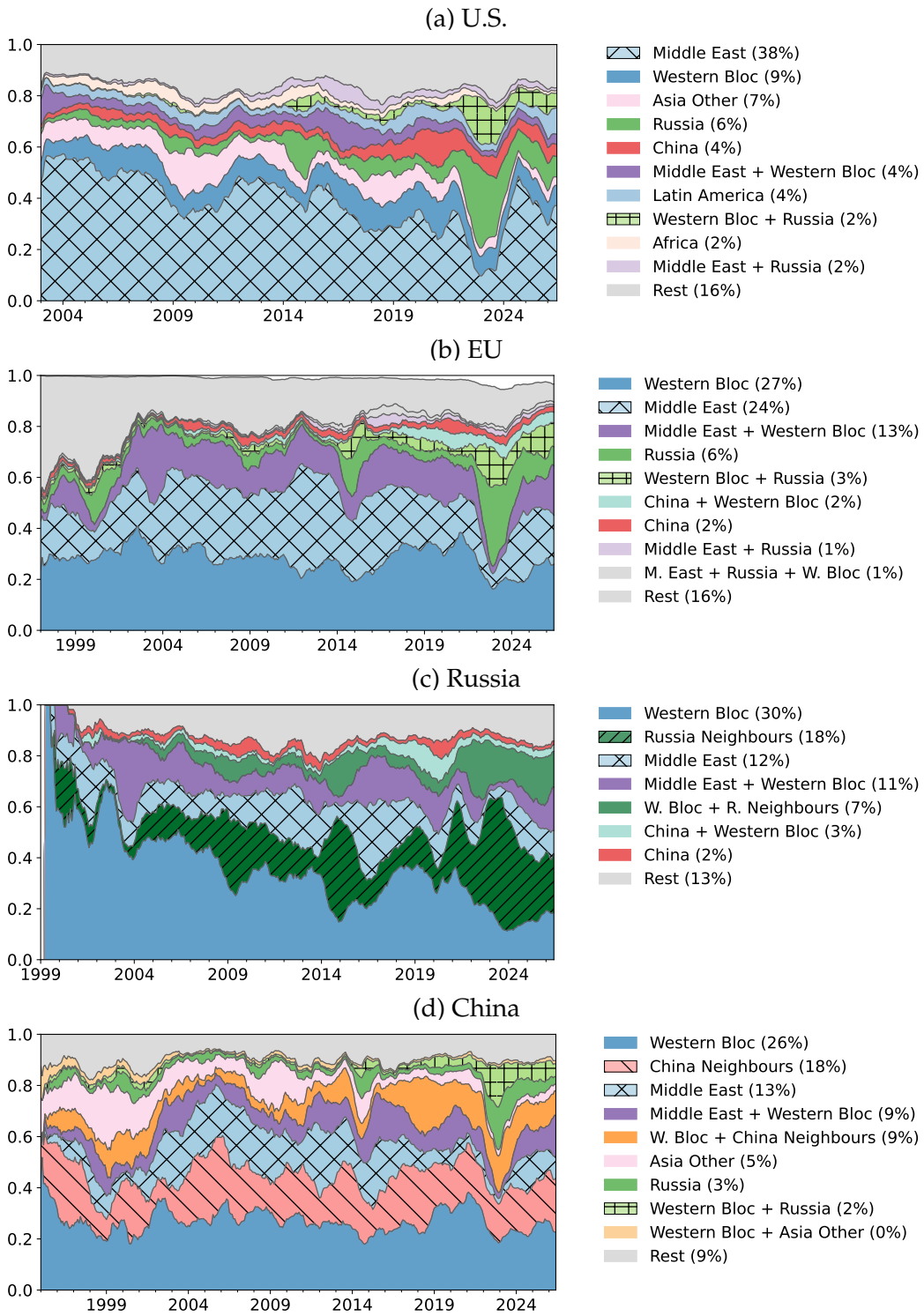
First, we split the world excluding the country of origin into a number of relevant regions - see Table F.1 for details. We then compose a query that forms an effective union of these regions, and compare the result with the “Other countries" query detailed in the section above. We then check whether our region combination gives us sufficient cover for this component - for Russia, the union of our regions gives a 98% coverage. If that number is too low, the implication is that there are other parts of the world not included in our selected regions that make up a substantial portion of the external GPR.

We now give reasons why we cannot simply increase the regional coverage for each country so that the components contribute to 100% of the external GPR. This is intrinsically linked to the way Factiva labels its articles and organises its searches. The first point to note is that Factiva´s “re" regions are nested, and multiple “re" classification levels are assigned to the same article. To give an example, the UK by definition forms part of “Europe", meaning that each article classified as the UK is also classified as Europe. Therefore, if we want to make sure the article includes at least one other part of the Europe than the UK, we would need to list these countries explicitly. The reason why that might become a problem brings us to the second point, which is that Factiva´s queries are length-limited. And since our approach involves logical combinations of various regions (see main text) - that, combined with the original GPR query, means that only a certain number of regions can be considered, each region being defined by a relatively short list of constituent parts. What is more, the query length is independent of the language, and given that German, for example, comes with a much longer GPR query, it is much more likely to exceed the query length for the same regional split than, say, the UK. That is the reason why for the purposes of this exercise we revise the baseline GPR query for Germany, and get rid of the Nuclear War component. This only gets rid of a small 2% of all the articles (we remind the reader the sample ends in 2023), so while we cannot be certain that it does not remove or add any peak, we are reasonably sure the part removed is statistically negligible.

In practical terms, for every decomposition we do, we end up with a 10-20% of external GPR being unclassified. This number is made up of two elements: 1. GPR components that focus entirely on regions outside of our focal interest set. For example, we know that Africa plays a large part in the French GPR narrative, and yet for the purpose of this article we did not include it in our focal regions list. Such articles focused entirely on Africa will contribute to the unclassified component. 2. A combination of selected focal regions that is too insignificant to count. When we compute our decomposition we order the region combinations by prevalence, and pick the top 8-10 components, which usually covers around 90% of the External GPR. The other combinations are subsumed under the unclassified category. Not that other undefined focal regions, when combined with these combinations, will also remain unclassified.

# G Decomposition: an alternative normalisation

Figure G.1: Decomposition of the External part of Geopolitical Risk




Notes: The top non-overlapping components of the external, outward-looking part of the GPR index. The values are monthly, with a rolling mean of one year. The constituents of each region, which depends on the country of origin, can be found detailed in Table F.1. Each plot area corresponds to the fraction of the external part of the GPR index limited to only the zone specified, with the exclusion of all the other zones detailed in Table F.1. To facilitate cross-country comparison, we show in brackets the average monthly fraction of External component starting from the year 2003.

# H The macroeconomic impact of GPR shocks: complete model results

## H.1 Macro data

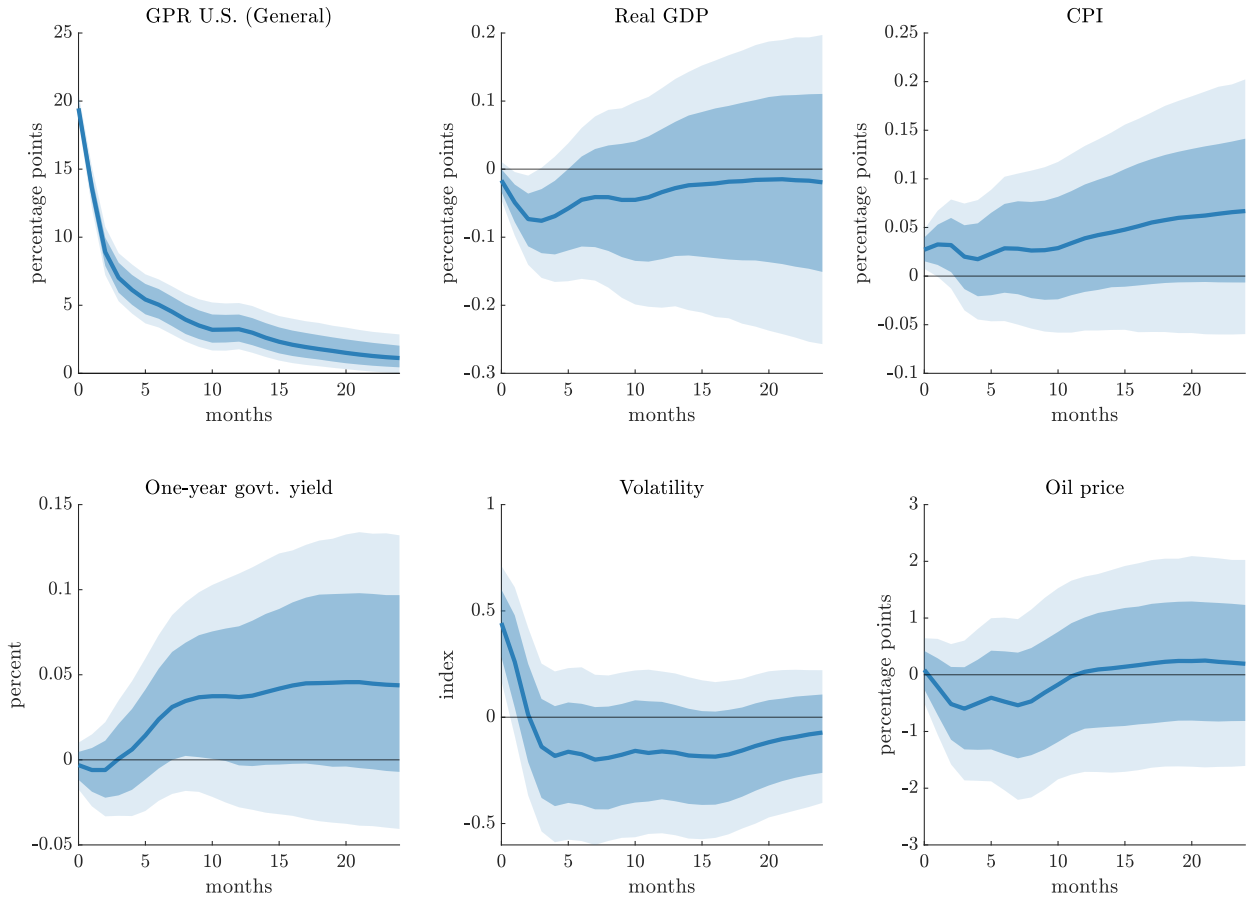
Table H.1: Description, Sources, and Coverage of Macroeconomic Data

| Variable | Description | Source | Sample |
|---|---|---|---|
| **United States** | | | |
| BBKMGDP | (Log cummulated) Brave-Butters-Kelley real gross domestic product | FRED | 1986M1–2026M6 |
| CPIAUCSL | Consumer price index | FRED | 1986M1–2026M6 |
| GS1 | One-year U.S. Treasury constant-maturity yield | FRED | 1986M1–2026M6 |
| VIXCLS | CBOE Volatility Index (VIX) | FRED | 1986M1–2026M6 |
| DCOILWTICO | West Texas Intermediate (WTI) crude oil price | FRED | 1986M1–2026M6 |
| **Russia** | | | |
| RSGDP...D | Real gross domestic product | Datastream | 2002Q1–2026Q2 |
| RSCONPRCF | Consumer price index | Datastream | 2002M1–2026M6 |
| RSPRATE. | Central Bank of Russia policy interest rate | Datastream | 2002M1–2026M6 |
| RSRTSIN | (Daily) RTS equity index | Datastream | 2002M1–2026M6 |
| OILURLB | Urals crude oil price | Datastream | 2002M1–2026M6 |
| **China** | | | |
| RealGDP | Real gross domestic product | Atlanta Fed | 1995M1–2026M6 |
| CPI | Consumer price index | Atlanta Fed | 1995M1–2026M6 |
| DepositRate1YBench | One-year benchmark deposit rate set by the People's Bank of China | Atlanta Fed | 1995M1–2026M6 |
| CHSCOMP | (Daily) Shanghai Stock Exchange Composite Index | Datastream | 1995M1–2026M6 |
| LCPCASH | London Metal Exchange (LME) copper price | Datastream | 1995M1–2026M6 |

*Notes:* For China and Russia, realized equity-market volatility is constructed as the sum of squared equity returns. Commodity prices are converted into real terms using the corresponding consumer price index. Variables available only at quarterly frequency are converted to monthly frequency using cubic-spline interpolation. Series are seasonally adjusted where appropriate.

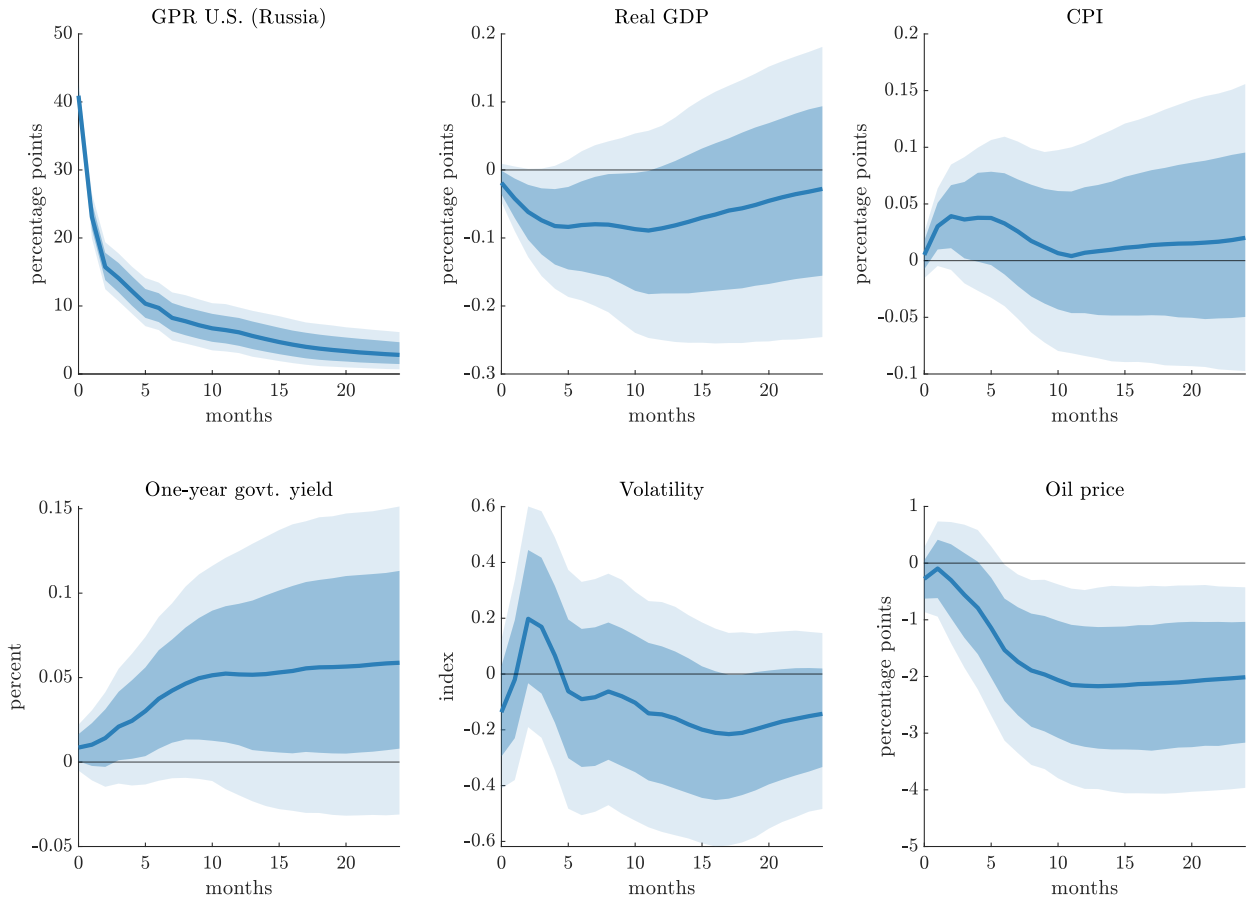
## H.2 Results for the U.S.

Figure H.2.1: Responses of the U.S. economy to an aggregate geopolitical risk shock



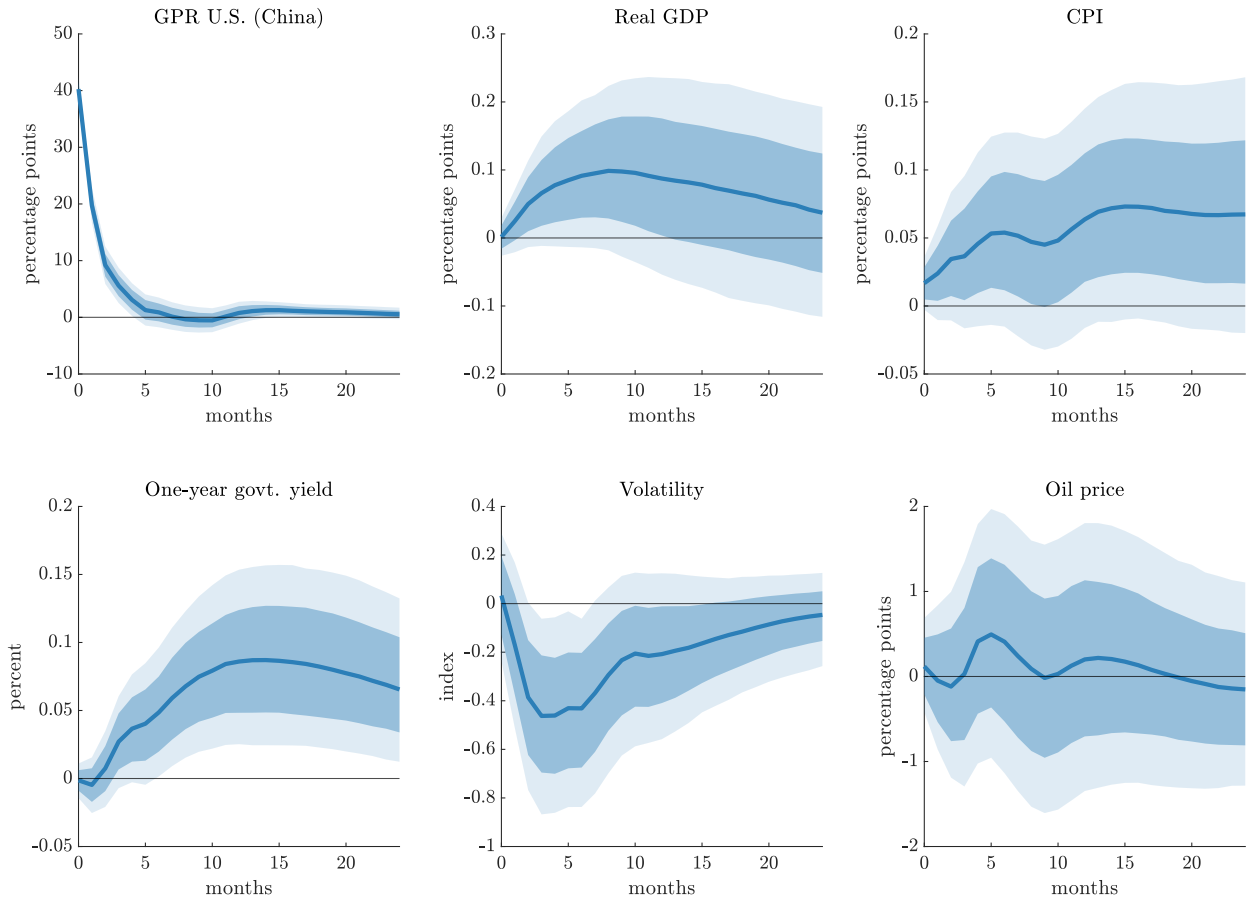

*Notes:* The figure reports impulse responses to a one-standard deviation increase in the U.S.'s aggregate GPR index, estimated using a Bayesian structural VAR over January 1986–June 2026. The model incorporates the COVID-19 volatility adjustment of Lenza and Primiceri (2022). The GPR index is ordered first in a recursive identification scheme, allowing geopolitical risk shocks to affect all other variables contemporaneously. Solid blue lines denote posterior median responses; darker and lighter shaded areas indicate 68% and 90% credible intervals, respectively. Units are indicated on the vertical axes, and the horizontal axes measure months after the shock.

Figure H.2.2: The impact of a bilateral GPR shock from Russia on the U.S. economy



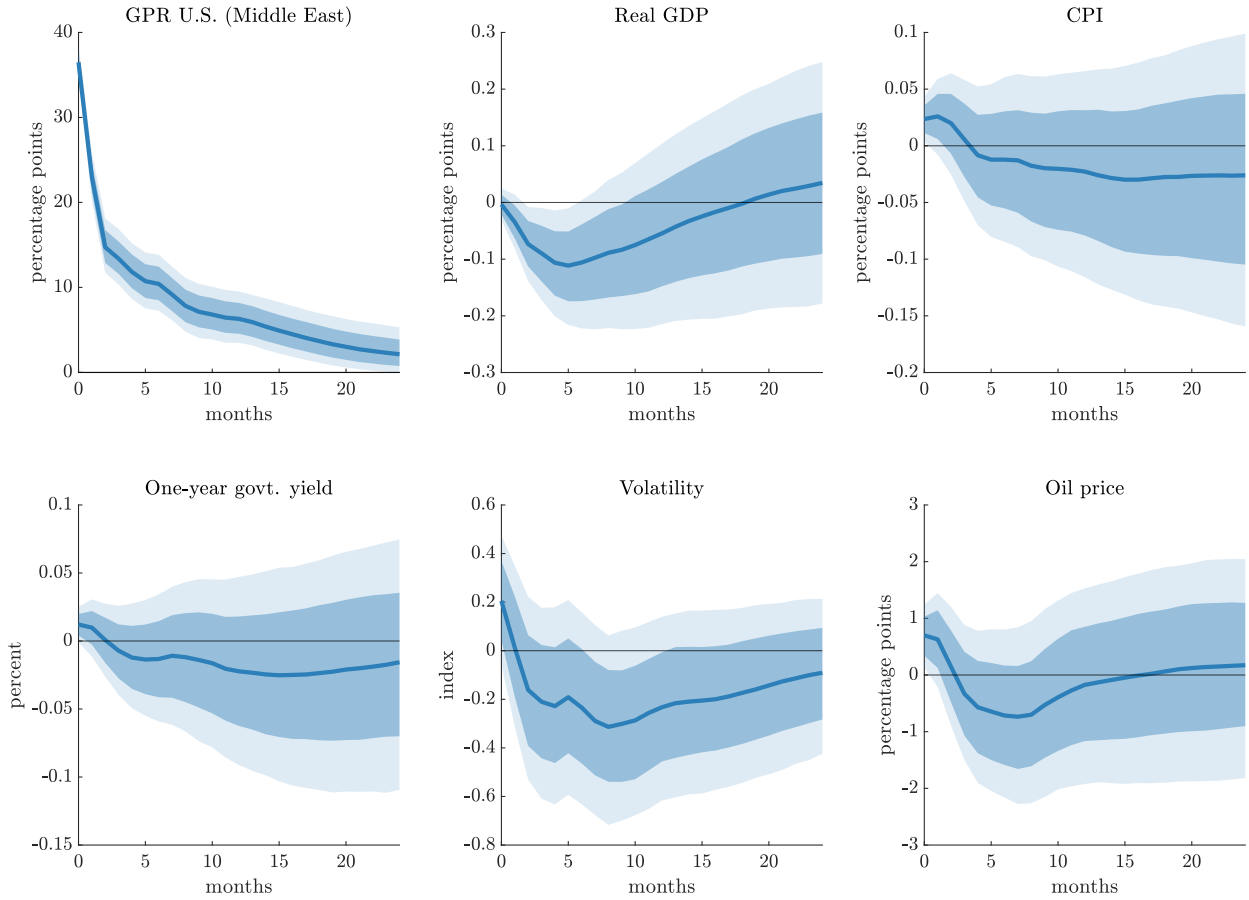

*Notes:* The figure reports impulse responses to a one-standard deviation increase in the selected U.S. bilateral GPR index, estimated using a Bayesian structural VAR over January 1986–June 2026. The model incorporates the COVID-19 volatility adjustment of Lenza and Primiceri (2022). The GPR index is ordered first in a recursive identification scheme, allowing geopolitical risk shocks to affect all other variables contemporaneously. Solid blue lines denote posterior median responses; darker and lighter shaded areas indicate 68% and 90% credible intervals, respectively. Units are indicated on the vertical axes, and the horizontal axes measure months after the shock.

Figure H.2.3: The impact of a bilateral GPR shock from China on the U.S. economy




*Notes:* The figure reports impulse responses to a one-standard deviation increase in the selected U.S. bilateral GPR index, estimated using a Bayesian structural VAR over January 1986–June 2026. The model incorporates the COVID-19 volatility adjustment of Lenza and Primiceri (2022). The GPR index is ordered first in a recursive identification scheme, allowing geopolitical risk shocks to affect all other variables contemporaneously. Solid blue lines denote posterior median responses; darker and lighter shaded areas indicate 68% and 90% credible intervals, respectively. Units are indicated on the vertical axes, and the horizontal axes measure months after the shock.

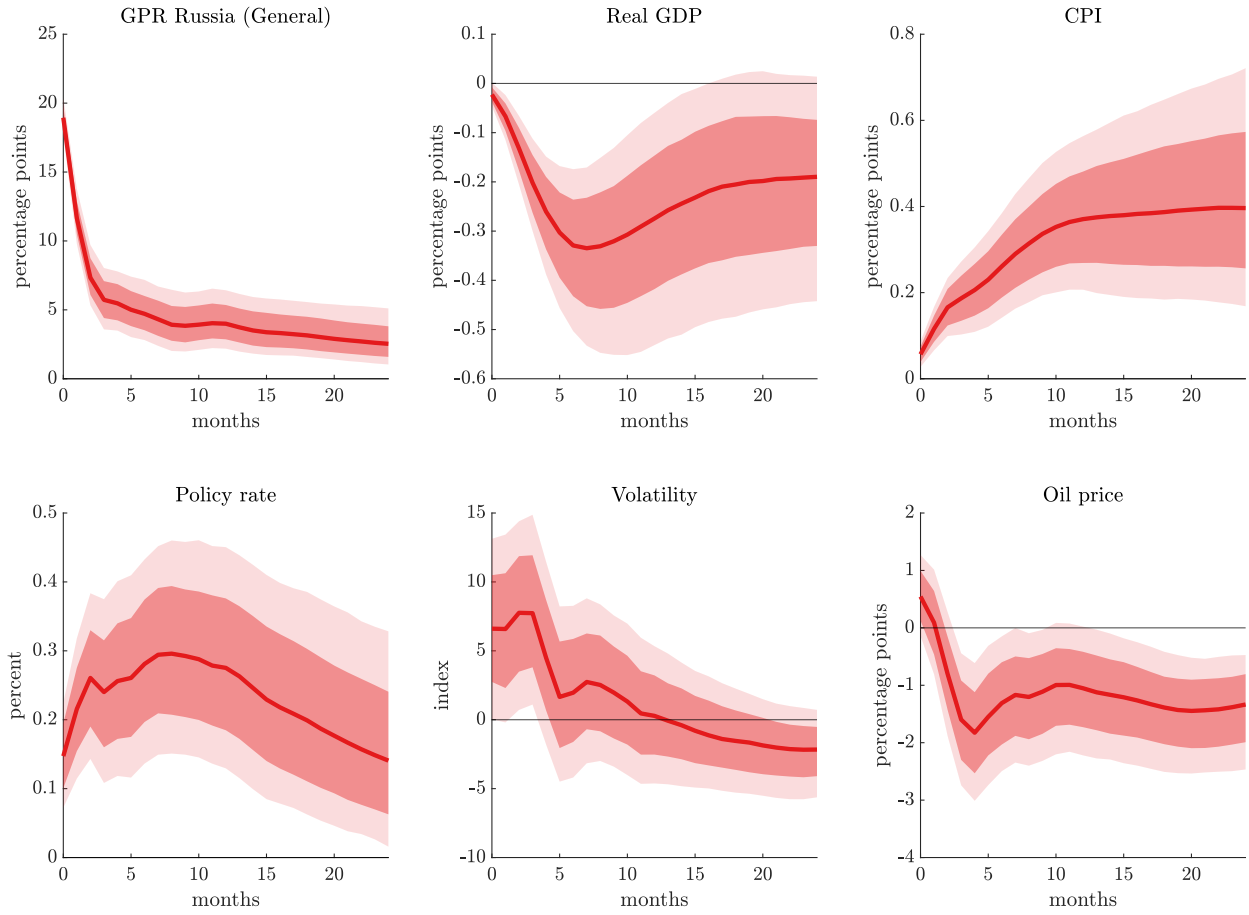
Figure H.2.4: The impact of a bilateral GPR shock from the Middle East on the U.S. economy




*Notes:* The figure reports impulse responses to a one-standard deviation increase in the selected U.S. bilateral GPR index, estimated using a Bayesian structural VAR over January 1986–June 2026. The model incorporates the COVID-19 volatility adjustment of Lenza and Primiceri (2022). The GPR index is ordered first in a recursive identification scheme, allowing geopolitical risk shocks to affect all other variables contemporaneously. Solid blue lines denote posterior median responses; darker and lighter shaded areas indicate 68% and 90% credible intervals, respectively. Units are indicated on the vertical axes, and the horizontal axes measure months after the shock.

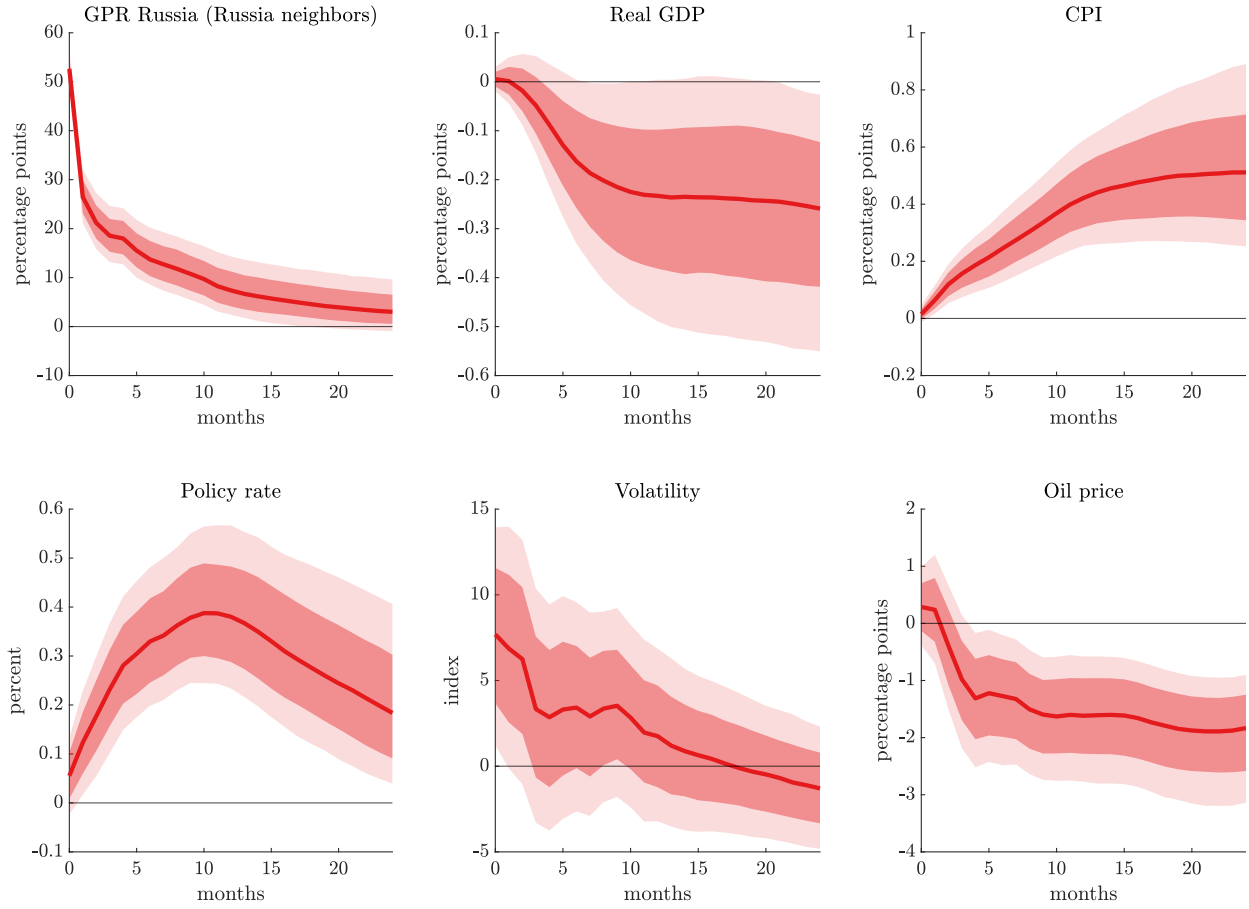
## H.3 Results for Russia

Figure H.3.5: Responses of the Russian economy to an aggregate geopolitical risk shock



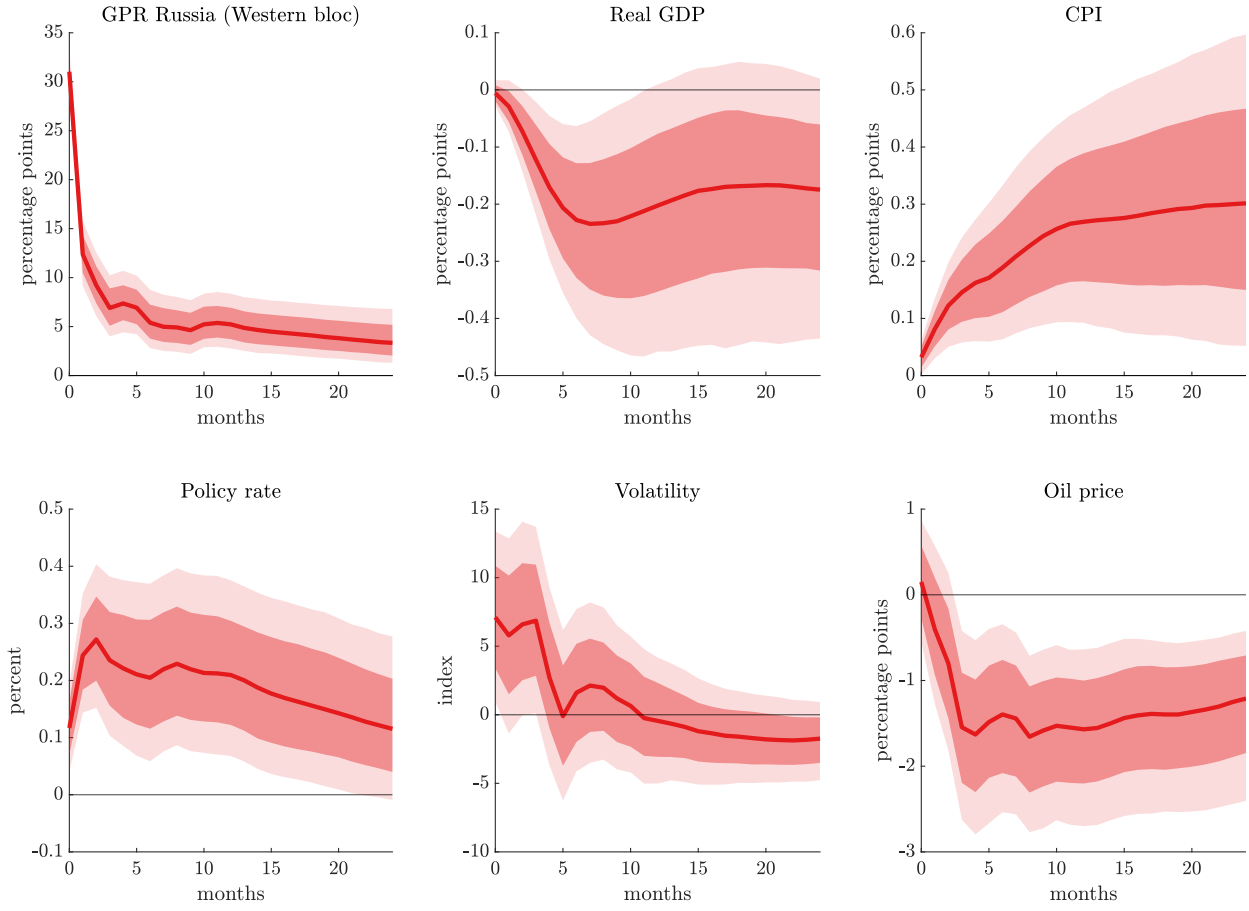

*Notes:* The figure reports impulse responses to a one-standard deviation increase in Russia's aggregate GPR index, estimated using a Bayesian structural VAR over January 2002–June 2026. The model incorporates the COVID-19 volatility adjustment of Lenza and Primiceri (2022). The GPR index is ordered first in a recursive identification scheme, allowing geopolitical risk shocks to affect all other variables contemporaneously. Solid red lines denote posterior median responses; darker and lighter shaded areas indicate 68% and 90% credible intervals, respectively. Units are indicated on the vertical axes, and the horizontal axes measure months after the shock.

Figure H.3.6: The impact of a bilateral GPR shock from Russia's neighbors on the Russian economy



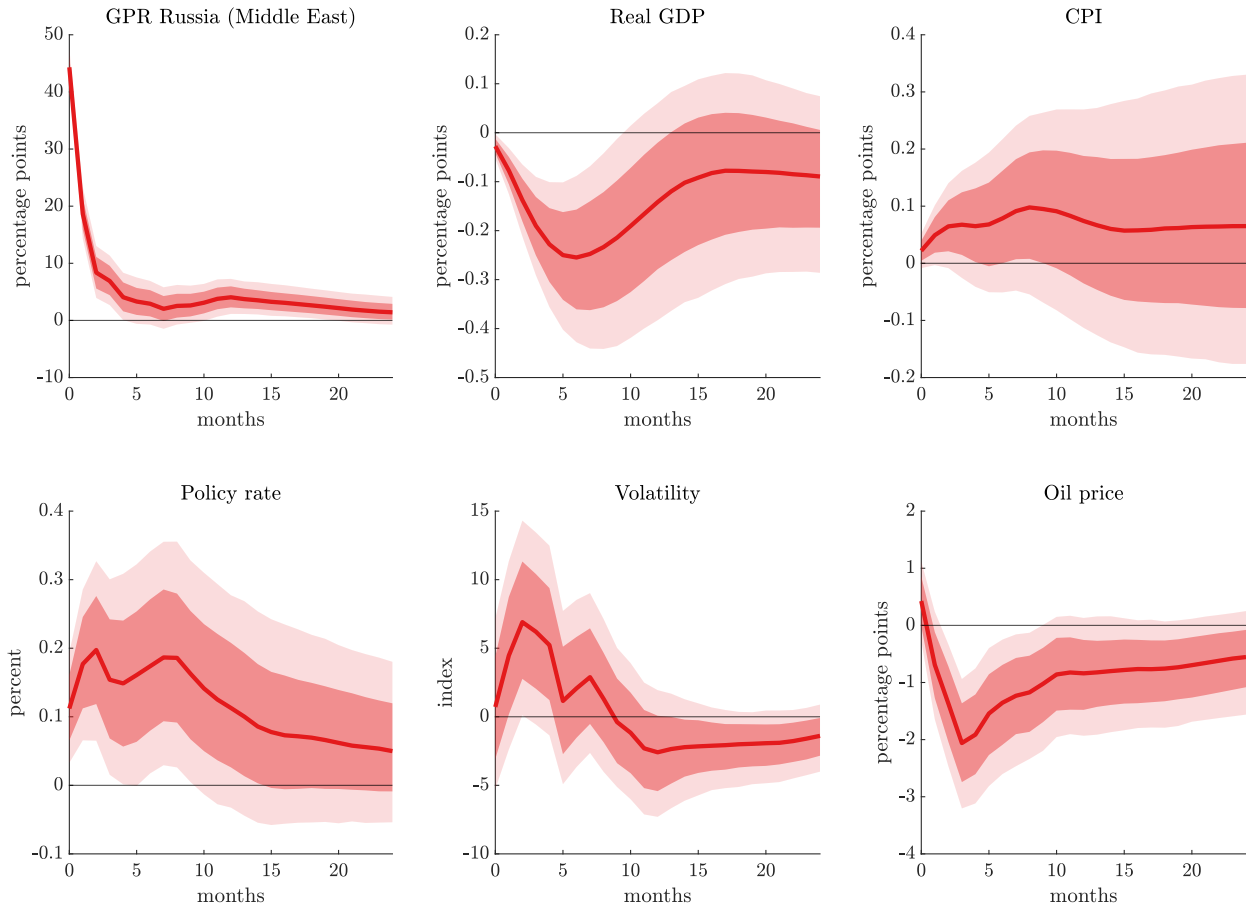

*Notes:* The figure reports impulse responses to a one-standard deviation increase in the selected Russian bilateral GPR index, estimated using a Bayesian structural VAR over January 2002–June 2026. The model incorporates the COVID-19 volatility adjustment of Lenza and Primiceri (2022). The GPR index is ordered first in a recursive identification scheme, allowing geopolitical risk shocks to affect all other variables contemporaneously. Solid red lines denote posterior median responses; darker and lighter shaded areas indicate 68% and 90% credible intervals, respectively. Units are indicated on the vertical axes, and the horizontal axes measure months after the shock.

Figure H.3.7: The impact of a bilateral GPR shock from the Western bloc on the Russian economy




*Notes:* The figure reports impulse responses to a one-standard deviation increase in the selected Russian bilateral GPR index, estimated using a Bayesian structural VAR over January 2002–June 2026. The model incorporates the COVID-19 volatility adjustment of Lenza and Primiceri (2022). The GPR index is ordered first in a recursive identification scheme, allowing geopolitical risk shocks to affect all other variables contemporaneously. Solid red lines denote posterior median responses; darker and lighter shaded areas indicate 68% and 90% credible intervals, respectively. Units are indicated on the vertical axes, and the horizontal axes measure months after the shock.

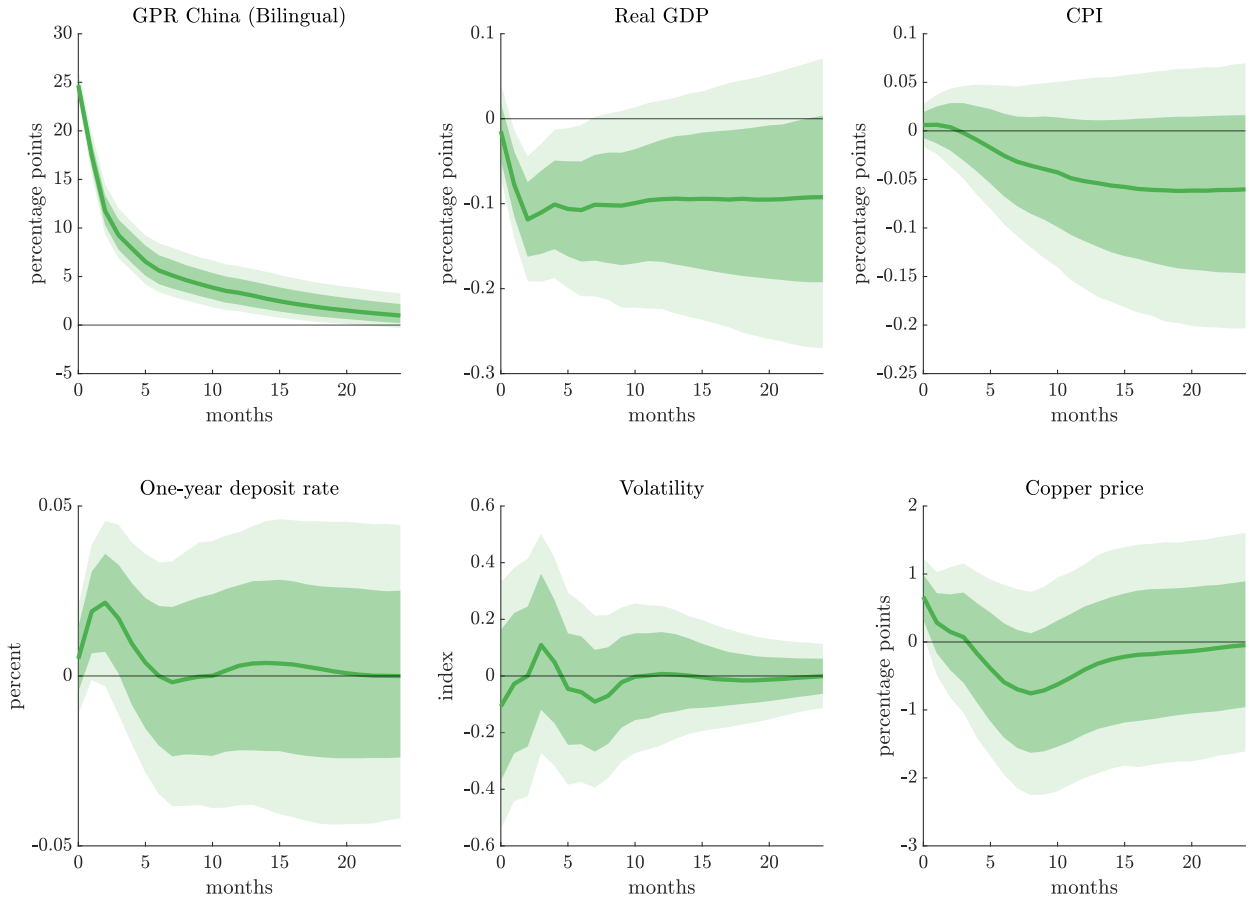
Figure H.3.8: The impact of a bilateral GPR shock from the Middle East on the Russian economy




*Notes:* The figure reports impulse responses to a one-standard deviation increase in the selected Russian bilateral GPR index, estimated using a Bayesian structural VAR over January 2002–June 2026. The model incorporates the COVID-19 volatility adjustment of Lenza and Primiceri (2022). The GPR index is ordered first in a recursive identification scheme, allowing geopolitical risk shocks to affect all other variables contemporaneously. Solid red lines denote posterior median responses; darker and lighter shaded areas indicate 68% and 90% credible intervals, respectively. Units are indicated on the vertical axes, and the horizontal axes measure months after the shock.

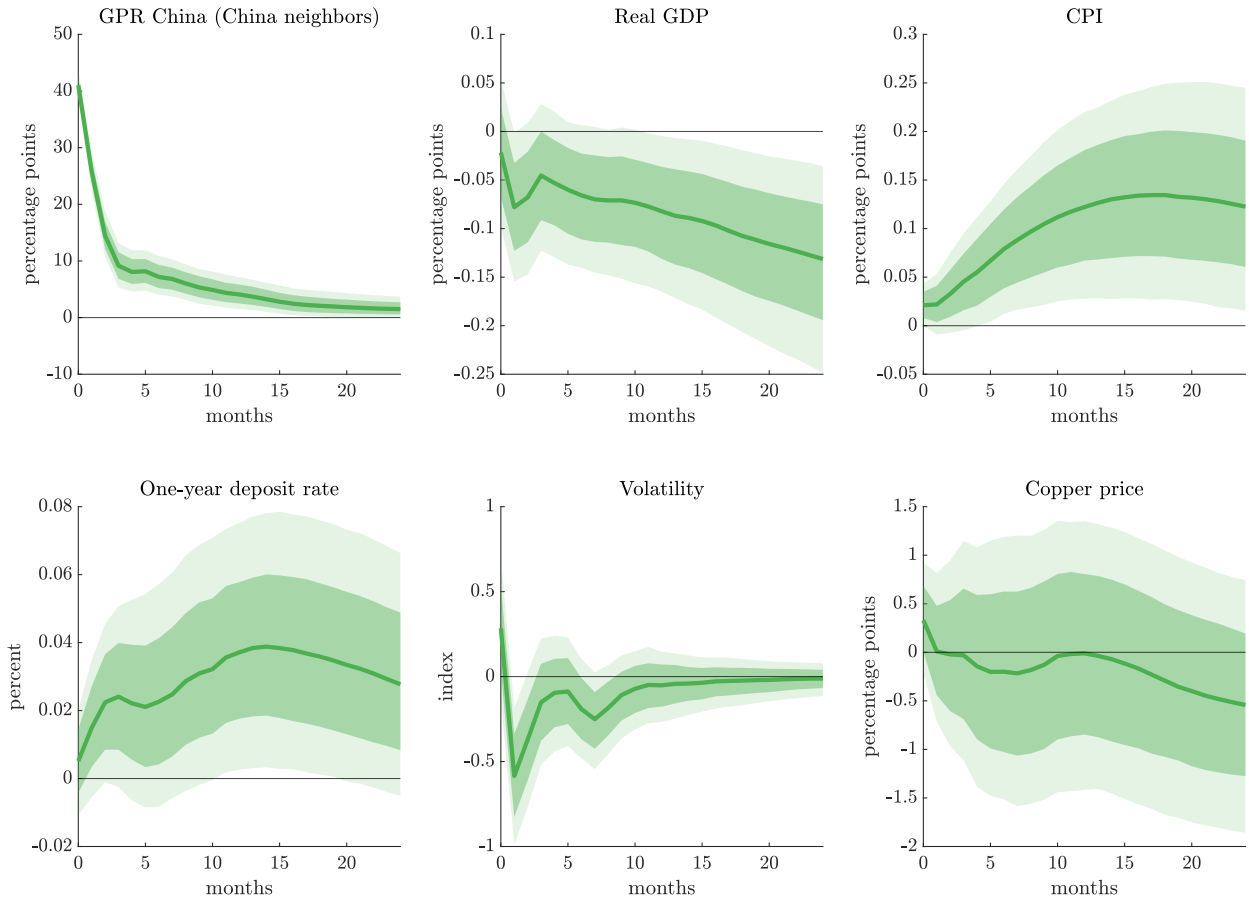
## H.4 Results for China

Figure H.4.9: Responses of the Chinese economy to an aggregate geopolitical risk shock



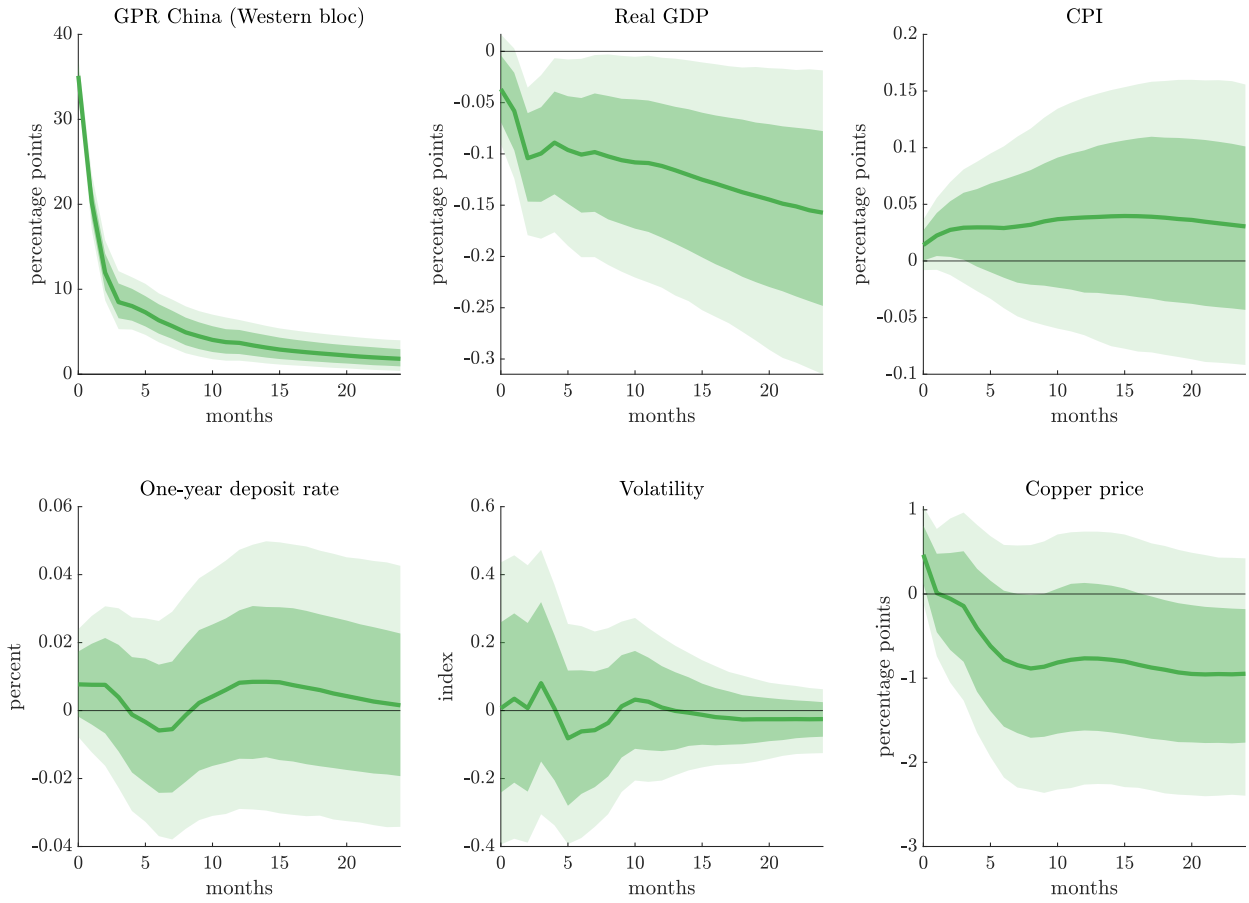

*Notes:* The figure reports impulse responses to a one-standard deviation increase in China's aggregate GPR index, estimated using a Bayesian structural VAR over January 1995–June 2026. The model incorporates the COVID-19 volatility adjustment of Lenza and Primiceri (2022). The GPR index is ordered first in a recursive identification scheme, allowing geopolitical risk shocks to affect all other variables contemporaneously. Solid red lines denote posterior median responses; darker and lighter shaded areas indicate 68% and 90% credible intervals, respectively. Units are indicated on the vertical axes, and the horizontal axes measure months after the shock.

Figure H.4.10: The impact of a bilateral GPR shock from China's neighbors on the Chinese economy



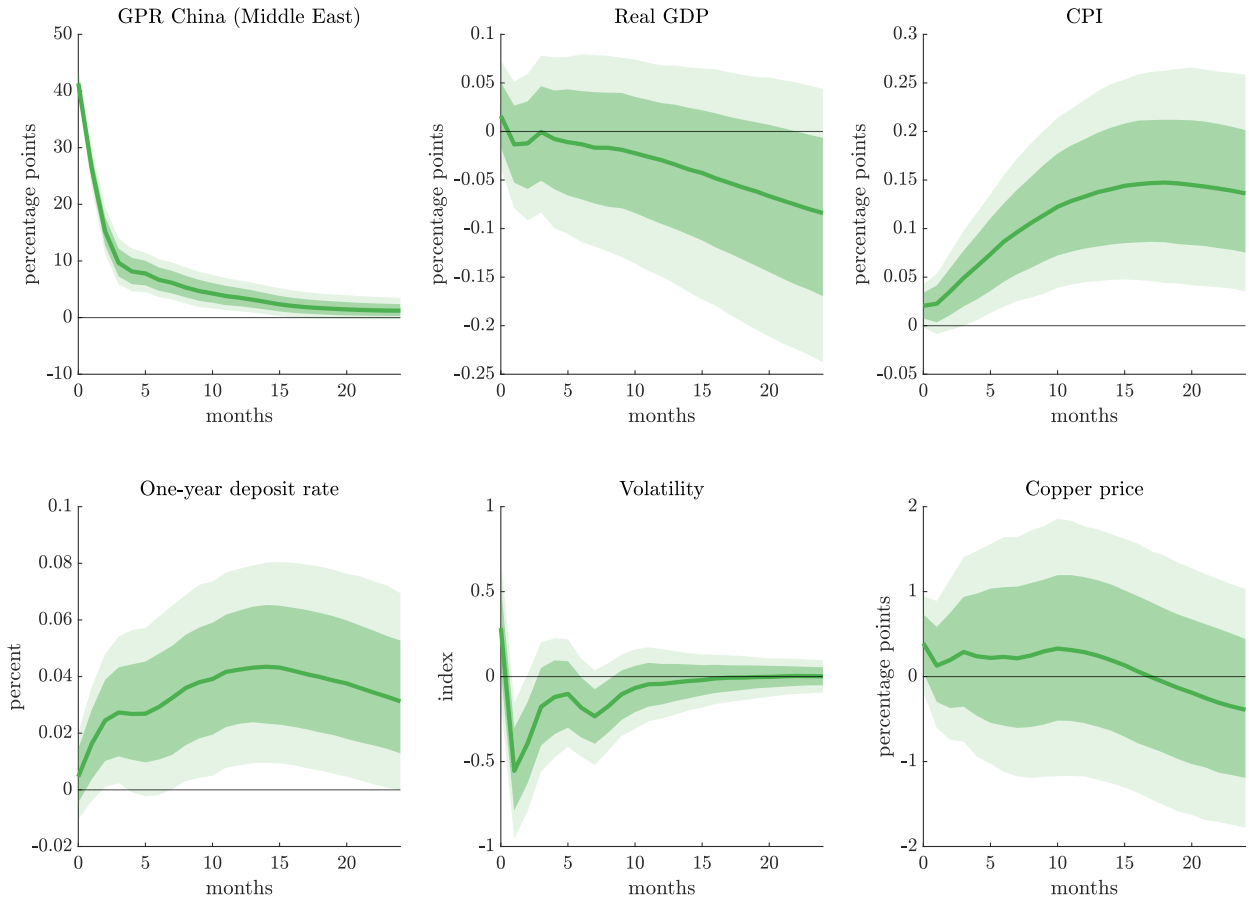

*Notes:* The figure reports impulse responses to a one-standard deviation increase in the selected Russian bilateral GPR index, estimated using a Bayesian structural VAR over January 1995–June 2026. The model incorporates the COVID-19 volatility adjustment of Lenza and Primiceri (2022). The GPR index is ordered first in a recursive identification scheme, allowing geopolitical risk shocks to affect all other variables contemporaneously. Solid red lines denote posterior median responses; darker and lighter shaded areas indicate 68% and 90% credible intervals, respectively. Units are indicated on the vertical axes, and the horizontal axes measure months after the shock.

Figure H.4.11: The impact of a bilateral GPR shock from the Western bloc on the Chinese economy




*Notes:* The figure reports impulse responses to a one-standard deviation increase in the selected Chinese bilateral GPR index, estimated using a Bayesian structural VAR over January 1995–June 2026. The model incorporates the COVID-19 volatility adjustment of Lenza and Primiceri (2022). The GPR index is ordered first in a recursive identification scheme, allowing geopolitical risk shocks to affect all other variables contemporaneously. Solid red lines denote posterior median responses; darker and lighter shaded areas indicate 68% and 90% credible intervals, respectively. Units are indicated on the vertical axes, and the horizontal axes measure months after the shock.

Figure H.4.12: The impact of a bilateral GPR shock from the Middle East on the Chinese economy




*Notes:* The figure reports impulse responses to a one-standard deviation increase in the selected Chinese bilateral GPR index, estimated using a Bayesian structural VAR over January 1995–June 2026. The model incorporates the COVID-19 volatility adjustment of Lenza and Primiceri (2022). The GPR index is ordered first in a recursive identification scheme, allowing geopolitical risk shocks to affect all other variables contemporaneously. Solid red lines denote posterior median responses; darker and lighter shaded areas indicate 68% and 90% credible intervals, respectively. Units are indicated on the vertical axes, and the horizontal axes measure months after the shock.